Nonlinear Learnable Triple-Domain Transform Tensor Nuclear Norm for Hyperspectral Image Super-Resolution

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***Abstract*— Tensor nuclear norm (TNN) has been widely employed as a regularization term for hyperspectral image super-resolution (HSISR). However, conventional TNN constraints based on discrete Fourier transform (DFT) often suffer from rank estimation biases and an inability to effectively capture complex spectral–spatial correlations, limiting their efficacy in HSISR. To address these challenges, we propose a nonlinear learnable triple (NLT)-domain transform framework that integrates a nonlinear transform, DFT, and self-learning adaptation. This multistage process promotes singular value concentration, improving low-rank approximation and rank estimation accuracy. Building upon this framework, we develop an NLT-transform-oriented tensor product, a truncated singular value decomposition (TSVD) operation, and a novel TNN (NLTN) tailored for HSISR. By incorporating spectral subspace estimation and clustering-based patch grouping, our approach effectively leverages spatial–spectral correlations and nonlocal self-similarities, leading to enhanced reconstruction quality. To further mitigate singular value over-penalization, we introduce a logarithmic-based generalized NLTNN (GNLTN) and formulate an optimization strategy based on the alternating direction method of multipliers (ADMMs). Extensive experiments demonstrate that our method significantly outperforms existing approaches in terms of fusion accuracy and visual fidelity, setting new benchmarks for HSISR. The code is available at https://github.com/xuhonghui96/GNLTN**

***Index Terms*— Alternating direction method of multipliers (ADMMs), discrete Fourier transform (DFT), hyperspectral image super-resolution (HSISR), low-rank approximation, nonlinear transform.**

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I. Introduction

HYPERSPECTRAL imaging (HSI) integrates spatial and spectral technologies to capture band reflectance data across a continuous range of wavelengths with narrow intervals [1], [2], [3], [4], [5], [6], [7]. Due to its distinctive ability to discriminate spectral features, HSI is widely applied in fields such as marine conservation [8] and target recognition [9], [10], [11]. However, HSI systems face a significant tradeoff between spatial and spectral resolutions. Specifically, to achieve an adequate signal-to-noise ratio across hundreds of narrow spectral bands, increasing the incident energy often requires sacrificing spatial resolution, leading to a decrease in texture details. Given the high costs and challenges associated with hardware upgrades, hyperspectral image super-resolution (HSISR) has emerged as a promising alternative to enhance the spatial size of the captured imagery. HSISR aims to fuse low-resolution hyperspectral images (LRHSIs) with corresponding high-resolution multispectral images (HRMSIs) of the same scene, thereby overcoming the limitations of traditional imaging systems.

In the past decade, HSISR has witnessed significant advancements, with numerous deep learning (DL)- and model-based methods emerging as the predominant approaches [12], [13], [14]. The powerful feature extraction capabilities of DL have led to exceptional performance across various image processing tasks, fueling growing interest in the application to image fusion. For instance, Zhu et al. [15] proposed the spectral correlation fusion network that effectively extracts spectral features, reduces band mismatches, integrates global contextual information, and emphasizes key regions. Hu et al. [16] introduced a novel HSISR network that maps the super-resolution task to the abundance domain and employs a spatial–spectral attention mechanism to refine low-resolution abundance estimates, effectively leveraging the overall information. Li et al. [17] proposed a spectral super-resolution framework that models cross-scale relationships, achieving satisfactory performance. Building on this work, they recently further introduced a coupling framework [18], which incorporates spectral unmixing into super-resolution and improves the final quality. Qi et al. [19] proposed a semi-supervised network for HSISR that combines supervised and unsupervised learning, requiring no extra training data or prior knowledge. Dong et al. [20] proposed BayeSR, a deep unfolding network

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that embeds Bayesian inference with Gaussian noise priors into its architecture, where the noise variance is interpreted as channel attention to explicitly encode image properties and enhance both performance and interpretability. Qu et al. [21] elaborated PMI-RFCoNet, a progressive multiiteration registration–fusion co-optimization network for unregistered HSI super-resolution, which jointly refines spatial alignment and spectral reconstruction through an iterative architecture incorporating deformable registration and adaptive spectral fusion modules. Despite their promising performance, several challenges persist. On the one hand, supervised networks often exhibit limited generalization ability, particularly when dealing with inconsistent test data, such as varying remote sensing sampling scenarios. On the other hand, the opaque model nature of DL models limits their interpretability, hindering the ability to fully understand and predict model outputs.

The model-based approach plays a pivotal role in effectively leveraging image priors and inferring the posterior of the latent HRHSI by probabilistically modeling the degradation process between the latent target and the observed LRHSI and HRMSI. Wang et al. [22] developed a Kronecker-based generalized tensor model to address arbitrary spatial degradations in HSI fusion, and proposed a block group sparsity-driven algorithm with provable recovery guarantees. Long et al. [23] proposed a parameter-free spectral–spatial optimization method for semi-informed HSI–MSI fusion, which enables efficient 64× high magnification without PSF estimation by leveraging matrix-based initialization, spectral error modeling, and spatial detail reconstruction. Among these approaches, the tensor nuclear norm (TNN) framework, leveraging tensor tubal rank and t-SVD [24], [25], [26], [27], [28], [29], enables effective high-order processing through tensor-tensor products implemented via DFT. This method mitigates the information loss typically associated with matricization or flattening operations, establishing TNN as a robust framework for low-rank representation. Building on this foundation, Dian et al. [1] extended the third-order TNN to handle arbitrary tensor modes, applying it to coefficient patches that simultaneously encode spatial, spectral, and nonlocal features, leading to significant fusion results. Bu et al. [3] further explored transferable subspace relationships between HRHSI and HRMSI, leveraging TNN to construct a nonlocal patch derived from multitensor low-rank subspaces. This strategy better captures the intricate structure of hyperspectral images. To more effectively capture the low-rank characteristics of tensors, researchers have explored various transform domains. Ma et al. [30] constructed an image fusion model based on the Hankel tensor subspace. After tensor decomposition, each factor component is mapped into a high-dimensional Hankel subspace via delay embedding, followed by TNN regularization to achieve fusion. Wu et al. [31] proposed a novel CD-based t-SVD framework and defined a corresponding tensor CD-rank based on a key structural insight. Leveraging this rank definition, they formulated a CD-rank-constrained minimization model for HSISR. These include linear and reversible domains such as the discrete cosine transform (DCT) [32], unitary [33], and gradient domains [34], as well as linear, nonreversible domains like Framelet [35] and dictionary [36] domains. Nonlinear and nonreversible transform [37] domains—such as those involving activation functions—have also been considered. Another category lies in the cascade domain. Note that most current approaches integrate DFT with other linear transforms [31], [38] or factor projection [39], [40], as well as the combination of gradient domains with other linear transforms [41]. The evolution of these transform-based TNN methods is summarized in Fig. 1.

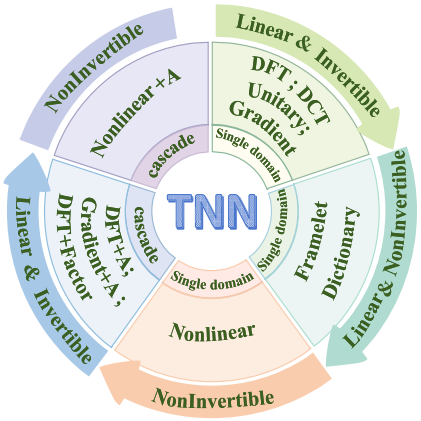
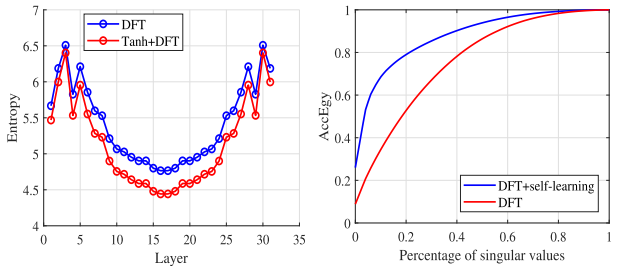


Fig. 1. Evolution of transform-based TNN methods. "+A" denotes linear and invertible transforms, while "nonlinear+A" represents our proposal, which plays as the latest model so that no progression arrow is given.

Within the TNN framework, the domain transforms are critical for capturing the intrinsic low-rank features, yet most previous efforts in HSISR have focused primarily on linear transforms. This narrow focus limits their ability to effectively model the nonlinear characteristics inherent in real-world hyperspectral data. Moreover, the rich and complex spectral profiles of hyperspectral datasets often challenge the use of a single and fixed transform domain. To date, the exploration of nonlinear coupling transforms remains limited, which may prevent full exploitation of the low-rank properties of hyperspectral images.

To achieve a more accurate low-rank approximation of the frontal slices of the transformed tensor, we introduce the nonlinear learnable triple (NLT)-domain concept. Building on this, we further develop an NLT-transform-oriented tensor product, a truncated singular value decomposition (TSVD) operation, and a TNN framework [novel TNN (NLTN)] tailored for the HSISR task. First, a nonlinear transform (Tanh) is applied to the original data, followed by the DFT on the transformed results. Fig. 2(a) compares the entropy distribution across different layers before and after the nonlinear transform, where entropy is computed as $H(i) = -\sum p(x_i) \log p(x_i)$, with $p(x_i)$ denoting the normalized magnitude of the transformed data at layer i. The results indicate that a standard DFT yields a more evenly distributed spectral representation. However, applying the nonlinear transform before DFT reduces entropy in intermediate layers, suppressing redundant information and enhancing the separability of low-rank structures, which serves as an effective prior for structured data representation. Building upon these transforms, we introduce a self-learning adaptive matrix, which is iteratively updated to refine the interaction between nonlinear and DFT transforms.

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(a) (b)

Fig. 2. Comparison of (a) entropy and (b) accuracy across different transform domains on the cd image from the CAVE dataset.

This matrix captures the global structure of the underlying tensor space, promoting self-learning and integrating dual-layer prior information. Fig. 2(b) demonstrates that combining DFT with self-learning requires fewer singular values to achieve the same cumulative energy ratio (AccEgy) compared to DFT alone, indicating that the singular value distribution in the triple-domain transform is more concentrated, facilitating the generation of low-rank slices. To further mitigate singular value over-penalization and reduce directional sensitivity, we apply logarithmic transforms to the three tensor modes and construct a logarithmic-based generalized NLTNN (GNLTN). In addition, we estimate spectral subspace coefficients and low-rank spectral bases of the LRHSI using SVD. To leverage shared spatial information between the HRMSI and HRHSI, we classify similar patches in the HRMSI through a clustering method, resulting in a low-rank 3-D tensor that captures both spectral low-rank properties and nonlocal similarities. The GNLTN is then applied to all distinct cubic groups derived from the subspace coefficients, reformulating the HSISR task as an optimization problem with GNLTN regularization, which is then solved using off-the-shelf optimization techniques.

Our contribution can be summarized as follows.

1) We investigate the potential of TSVD within a triple-domain transform framework, consisting of nonlinear, DFT, and self-learning transforms. In doing so, we introduce key concepts, such as t-product and tubal rank, which significantly enhance existing TSVD theory. To the best of our knowledge, this work presents the first nonlinear-cascade-transform-based TSVD framework.
2) We theoretically derive a logarithmic-based generalized NLTN as a convex relaxation of the tensor NLT-transform rank. By applying it to the three permutation forms of the tensor obtained through clustering, the concerned spatial low-rank, spectral low-rank, and nonlocal features can be simultaneously explored.
3) We elaborate on an effective alternating direction method of multipliers (ADMMs) to address the optimization challenge of the proposed model. Numerical results demonstrate that our approach outperforms most existing methods in terms of fusion performance.

II. Notations and Preliminaries

A. Notations

In this work, we use specific notations to represent scalars, vectors, matrices, and tensors, denoted as x, $\boldsymbol{x}$, $\boldsymbol{X}$, and $\mathcal{X}$, respectively. The (i, j, k)th entry of $\mathcal{X}$ is denoted as $\mathcal{X}(i, j, k)$ or $x_{i,j,k}$. The mode-k unfolding (matricization) of $\mathcal{X}$ is denoted as $\boldsymbol{X}_{(k)}$. The mode-k product between $\mathcal{X}$ and a matrix $\boldsymbol{U} \in \mathbb{R}^{J\times I_k}$ is denoted by $\mathcal{X} \times_k \boldsymbol{U}$, which produces a third-order tensor of size $I_1 \times I_2 \times I_3$, and satisfies the relation $(\mathcal{X} \times_k \boldsymbol{U})(k) = \boldsymbol{U} \cdot \boldsymbol{X}_{(k)}$ for $k = 1, 2, 3$. In this article, we apply mode-1, mode-2, and mode-3 products in a consistent manner based on this definition. For reference, we denote the horizontal, lateral, and frontal slices of $\mathcal{X}$ as $\mathcal{X}(i, :, :)$, $\mathcal{X}(:, i, :)$, and $\mathcal{X}(:, :, i)$, respectively. The discrete Fourier transform (DFT) along mode-3 is defined as $\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [\,], 3) = \mathcal{X} \times_3 \boldsymbol{F}$, where $\boldsymbol{F}$ is the DFT matrix. The inner product between two third-order tensors $\mathcal{X}$ and $\mathcal{Y}$ is defined as $\langle \mathcal{X}, \mathcal{Y} \rangle := \sum i, j, k x_{i,j,k} y_{i,j,k}$.

B. Problem Formulation

In this work, our objective is to reconstruct the HRHSI, denoted as $\mathcal{X} \in \mathbb{R}^{W\times H\times S}$, by fusing two complementary data sources: the LRHSI $\mathcal{Y} \in \mathbb{R}^{w\times h\times S}$ and the HRMSI $\mathcal{Z} \in \mathbb{R}^{W\times H\times s}$ captured from the same scene. While HRHSI provides both high spatial and spectral fidelity, LRHSI and HRMSI only capture partial information. Specifically, $\mathcal{Y}$ retains spectral details but at reduced spatial resolution ($w < W$ and $h < H$), whereas $\mathcal{Z}$ preserves spatial resolution but offers fewer spectral bands ($s < S$).

Mathematically, the LRHSI can be modeled as a spatially downsampled version of HRHSI

$$\boldsymbol{Y}_{(3)} = \boldsymbol{X}_{(3)}\boldsymbol{B}\boldsymbol{S} \tag{1}$$

where $\boldsymbol{Y}_{(3)} \in \mathbb{R}^{S\times \text{hw}}$ and $\boldsymbol{X}_{(3)} \in \mathbb{R}^{S\times \text{HW}}$ are the mode-3 unfoldings of $\mathcal{Y}$ and $\mathcal{X}$. Here, $\boldsymbol{S} \in \mathbb{R}^{\text{HW}\times \text{hw}}$ performs spatial downsampling, while $\boldsymbol{B} \in \mathbb{R}^{\text{HW}\times \text{HW}}$ models convolution blur, represented in the Fourier domain as

$$\boldsymbol{B} = \boldsymbol{F}_1 \boldsymbol{K} \boldsymbol{F}_1^{-1} \tag{2}$$

where $\boldsymbol{K}$ is the diagonal matrix of eigenvalues, and $\boldsymbol{F}_1$ represents the Fourier transform. Similarly, HRMSI is derived by spectrally downsampling HRHSI, such as

$$\boldsymbol{Z}_{(3)} = \boldsymbol{R}\boldsymbol{X}_{(3)} \tag{3}$$

where $\boldsymbol{Z}_{(3)} \in \mathbb{R}^{s\times \text{HW}}$ is the mode-3 unfolding of $\mathcal{Z}$, and $\boldsymbol{R} \in \mathbb{R}^{s\times S}$ represents the spectral response function.

To address the high correlation among hyperspectral bands, HRHSI is assumed to exhibit a low-rank structure in the spectral domain. As shown in [42], this property can be captured through subspace decomposition

$$\boldsymbol{X}_{(3)} = \boldsymbol{D}\boldsymbol{A}_{(3)} \tag{4}$$

where $\boldsymbol{D} \in \mathbb{R}^{S\times L}$ defines the spectral basis, and $\boldsymbol{A}_{(3)} \in \mathbb{R}^{L\times \text{HW}}$ represents the coefficients in the reduced-dimensional space. By incorporating this decomposition, the spatial and spectral degradation processes are reformulated as

$$\boldsymbol{Y}_{(3)} = \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}, \quad \boldsymbol{Z}_{(3)} = \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}. \tag{5}$$

Since both HRHSI and LRHSI share identical spectral properties, they are governed by the same spectral basis $\boldsymbol{D}$. Consequently, the problem can be reduced to optimizing the

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coefficients in the reduced-dimensional space. This allows HSISR to be reformulated as solving for $\boldsymbol{A}_{(3)}$, as follows:

$$\min_{\boldsymbol{A}_{(3)}} \left\| \boldsymbol{Y}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S} \right\|_F^2 + \left\| \boldsymbol{Z}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)} \right\|_F^2 + \lambda R\left(\boldsymbol{A}_{(3)}\right) \tag{6}$$

where $R(\cdot)$ introduces regularization, and λ balances the tradeoff between data fidelity and prior constraints.

III. NLT-Transform TNN

Given any tensor $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, owing to the circular convolution, its t-SVD can be computed via the summation of the matrix nuclear norm of $(\mathcal{L} \times_3 \boldsymbol{F})$' s frontal slices. Instead of using the specific DFT, we introduce a new definition of NLT-transform-based TSVD decomposition. This framework allows for the intuitive derivation of algebraic operations associated with 3-D tensors, such as the NLT-oriented t-product, t-SVD, and low-rank approximation.

Definition 1 (NLT Transform): Let $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ be a tensor with element-wise nonlinear activation function ϕ, the DFT matrix $\boldsymbol{F}$, and the self-learning transform matrix $\boldsymbol{U}$, obtained by applying dynamic SVD to the mode-3 unfolding of the target tensor, i.e., $[\boldsymbol{U}, \ldots] = \text{SVD}(\boldsymbol{L}_{(3)})$. The NLT transform $N : \mathbb{R}^{I_1 \times I_2 \times I_3} \to \mathbb{C}^{I_1 \times I_2 \times I_3}$ is defined as the composition of a nonlinear operation $N_1 : \mathbb{R}^{I_1 \times I_2 \times I_3} \to \mathbb{R}^{I_1 \times I_2 \times I_3}$ and a linear transform $N_2 : \mathbb{R}^{I_1 \times I_2 \times I_3} \to \mathbb{C}^{I_1 \times I_2 \times I_3}$, i.e.,

$$N := N_1 \circ N_2 := \phi \circ (\boldsymbol{F} \circ \boldsymbol{U}) := \phi \circ \boldsymbol{C} \tag{7}$$

where $\boldsymbol{C} = \boldsymbol{F} \circ \boldsymbol{U}$ represents the composite linear transform matrix. The NLT transform is applied to $\mathcal{L}$ along the third dimension, and can be expressed as:

$$\begin{aligned} N(\mathcal{L}) &= N_2(N_1(\mathcal{L})) = \mathcal{L}_\phi \times_3 (\boldsymbol{F} \times \boldsymbol{U}) \\ &= \text{fold}_3\left(\phi\left(\boldsymbol{L}_{(3)}\right)\right) \times_3 \boldsymbol{C} \end{aligned} \tag{8}$$

where $\mathcal{L}_\phi = N_1(\mathcal{L})$ denotes the nonlinear transform of $\mathcal{L}$ along its third mode, and the operation fold_3 refers to the folding of a tensor along its third mode.

Remark 1: Given that $\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{I}$ and $\boldsymbol{U}\boldsymbol{U}^{\mathrm{T}} = \boldsymbol{I}$, the combination $\boldsymbol{F} \cdot \boldsymbol{U}$ represents a semi-orthogonal linear transform. Thus, the proposed NLT transform can be interpreted as a single-layer neural network that combines both linear and nonlinear components. Specifically, the linear transform corresponds to a semi-orthogonal fully connected layer, while the nonlinear transform is governed by the activation function. Note that semi-orthogonality has been shown to enhance the training stability and performance of deep convolutional neural networks, highlighting the potential benefits of this approach [43].

Remark 2: Our self-learning transformation is a form that depends on the data rather than a fixed pattern. From an algorithmic perspective, unlike the data-independent random projection, our semi-orthogonal transform is embedded in the model iteration process, which is updated by $[\boldsymbol{U}, \ldots] = \text{SVD}(\boldsymbol{L}_{(3)})$ at each iteration. Due to the data restoration during iterations, the learned transformation matrix obtained through SVD gradually approaches the data eigenvalues, dynamically achieving low-rank approximation in the transformation process.

Building on the previously observed phenomena, we proceed to define the key properties of the proposed NLT transform t-product, which are critical for subsequent analysis.

Definition 2 (NLT transform t-product): For any two given tensors $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and $\mathcal{K} \in \mathbb{R}^{I_2 \times I_4 \times I_3}$, with nonlinear activation function N_1 and linear transform matrix $N_2 : \boldsymbol{C} = \boldsymbol{F} \times \boldsymbol{U}$, the NLT transform t-product $\mathcal{J}$ of $\mathcal{K}$ and $\mathcal{L}$ is denoted as a tensor with size $I_1 \times I_4 \times I_3$, i.e.,

$$\begin{aligned} &N_1(\mathcal{J}) = N_1(\mathcal{L} *_{\boldsymbol{C}} \mathcal{K}) = \boldsymbol{C}^{-1}(N_2(N_1(\mathcal{L}) \odot N_1(\mathcal{K}))) \\ &\Longleftrightarrow \mathcal{J}_\phi = \mathcal{L}_\phi *_{\boldsymbol{C}} \mathcal{K}_\phi = \boldsymbol{C}^{-1}\left(\bar{\mathcal{L}}_\phi \odot \bar{\mathcal{K}}_\phi\right) \end{aligned} \tag{9}$$

where $\bar{\mathcal{L}}_\phi = N_2(\mathcal{L}_\phi) = \mathcal{L}_\phi \times_3 \boldsymbol{C}$ and $\odot$ is the frontal-slicewise product.

Remark 3: Due to the lack of semi-orthogonal properties in the nonlinear transform, any given tensor $\mathcal{L}$ will be defined and analyzed based on the output tensor $N_1(\mathcal{L})$ derived from the first-stage invariant transform.

The proposed t-product under the NLT transform, akin to the original t-product, can also be represented as a matrix product, that is,

$$\mathcal{J}_\phi = \mathcal{L}_\phi *_{\boldsymbol{C}} \mathcal{K}_\phi \Longleftrightarrow \bar{\boldsymbol{J}}_\phi = \bar{\boldsymbol{L}}_{\boldsymbol{\phi}} \bar{\boldsymbol{K}}_{\boldsymbol{\phi}}. \tag{10}$$

For instance, the terms $\bar{\boldsymbol{L}}_{\boldsymbol{\phi}}$ is as a block diagonal matrix with its ith block on the diagonal as the ith frontal slice $\bar{\boldsymbol{L}}_{\boldsymbol{\phi}}^{(i)}$ of $\bar{\mathcal{L}}_\phi$, i.e.,

$$\bar{\boldsymbol{L}}_{\boldsymbol{\phi}} = \text{bdiag}(\bar{\mathcal{L}}_\phi) = \begin{bmatrix} \bar{\boldsymbol{L}}_{\boldsymbol{\phi}}^{(1)} & & & \\ & \bar{\boldsymbol{L}}_{\boldsymbol{\phi}}^{(2)} & & \\ & & \ddots & \\ & & & \bar{\boldsymbol{L}}_{\boldsymbol{\phi}}^{(I_3)} \end{bmatrix} \tag{11}$$

where bdiag is an operator mapping tensor $\bar{\mathcal{L}}_\phi$ to the block diagonal matrix $\bar{\boldsymbol{L}}_{\boldsymbol{\phi}}$.

Definition 3 (NLT Transform Identity Tensor): Given the nonlinear activation function N_1 and linear transform N_2, the $N_1(\mathcal{I}) = \mathcal{I}_\phi \in \mathbb{R}^{I_1 \times I_1 \times I_3}$ is called the identity tensor if all frontal slices of $N_2(N_1(\mathcal{I})) = \bar{\mathcal{I}}_\phi$ are $I_1 \times I_1$ identity matrices. In this case, $\mathcal{I}$ is referred to as the NLT transform identity tensor.

Definition 4 (NLT Transform Tensor Transpose): Given a linear transform N_2 and a nonlinear transform that maps $\mathcal{L}$ to $\mathcal{L}_\phi$, the tensor transpose of $\mathcal{L}_\phi$, denoted by $\mathcal{L}_\phi^{\mathrm{T}}$, is defined as $N_2(\mathcal{L}_\phi)(:, :, i) = (\bar{\mathcal{L}}_\phi(:, :, i))^{\mathrm{T}}$, for $i = 1, \ldots, I_3$.

Definition 5 (NLT Transform Orthogonal Tensor): Let $\mathcal{L}$ be a tensor transformed to $\mathcal{L}_\phi$ under a nonlinear transform. If $\mathcal{L}_\phi^{\mathrm{T}} *_{\boldsymbol{C}} \mathcal{L}_\phi = \mathcal{L}_\phi *_{\boldsymbol{C}} \mathcal{L}_\phi^{\mathrm{T}} = \mathcal{I}$, then $\mathcal{L}_\phi$ is an orthogonal tensor, and $\mathcal{L}$ is called the NLT transform orthogonal matrix.

Definition 6 (NLT Transform TSVD): Given a nonlinear transform that maps $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ to $\mathcal{L}_\phi \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the NLT transform TSVD (NTSVD) is defined as

$$\mathcal{L}_\phi = \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{S}_\phi *_{\boldsymbol{C}} \mathcal{V}_\phi^{\mathrm{T}} \tag{12}$$

where $\mathcal{U}_\phi \in \mathbb{R}^{I_1 \times I_1 \times I_3}$ and $\mathcal{V}_\phi \in \mathbb{R}^{I_2 \times I_2 \times I_3}$ are NLT transform orthogonal tensors, and $\mathcal{S}_\phi \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is an f-diagonal tensor (i.e., the tensor with diagonal elements along the f-diagonal).

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Definition 7 (NLT Transform Tensor Tubal Rank): Given any $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and its nonlinear transform product $\mathcal{L}_\phi \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, let $\mathcal{S}_\phi$ be from the $\mathcal{L}_\phi = \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{S}_\phi *_{\boldsymbol{C}} \mathcal{V}_\phi^{\mathrm{T}}$, the NTSVD rank of $\mathcal{L}$ is defined as

$$\operatorname{rank}_{\mathrm{NTSVD}}(\mathcal{L}) := \sharp\{i, \mathcal{S}_\phi(i,i,:) \neq \mathbf{0}\} \tag{13}$$

where $\sharp$ denotes the cardinality of a set.

Building on [31], the NLT transform TNN is derived from the spectral norm, as defined below.

Definition 8 (NLT Transform Tensor Spectral Norm): Let $\mathcal{L}$ be a tensor transformed to $\mathcal{L}_\phi$ under the nonlinear transform. The NLT transform tensor spectral norm of $\mathcal{L}$ is defined as $\|\mathcal{L}_\phi\| := \|\bar{\boldsymbol{L}}_\phi\|$.

To construct the nuclear norm, we assume the composite linear transform $\boldsymbol{C}$ satisfies

$$\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C} = \boldsymbol{C}\boldsymbol{C}^{\mathrm{T}} \Longleftrightarrow (\boldsymbol{F}\boldsymbol{U})^{\mathrm{T}}(\boldsymbol{F}\boldsymbol{U}) = (\boldsymbol{F}\boldsymbol{U})(\boldsymbol{F}\boldsymbol{U})^{\mathrm{T}} = \ell_1\ell_2\boldsymbol{I} \tag{14}$$

where ℓ_1 and ℓ_2 are scalar constants. This assumption ensures isotropic properties in the transformed space, facilitating the computation of norms. Under this assumption, the inner product and Frobenius norm satisfy

$$\langle \mathcal{L}_\phi, \mathcal{K}_\phi \rangle = \frac{1}{\ell_1\ell_2}\langle \bar{\boldsymbol{L}}_\phi, \bar{\boldsymbol{K}}_\phi \rangle \tag{15}$$

and

$$\|\mathcal{L}_\phi\|_F = \frac{1}{\ell_1\ell_2}\|\bar{\boldsymbol{L}}_\phi\|_F. \tag{16}$$

Subsequently, given any $\mathcal{K} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and $\boldsymbol{J} \in \mathbb{R}^{I_1 I_3 \times I_2 I_3}$, we have

$$\|\mathcal{L}_\phi\|_{*,\boldsymbol{C}} := \sup_{\|\mathcal{K}_\phi\| \le 1} \langle \mathcal{L}_\phi, \mathcal{K}_\phi \rangle = \sup_{\|\bar{\boldsymbol{K}}_\phi\| \le 1} \frac{1}{\ell_1\ell_2}\langle \bar{\boldsymbol{L}}_\phi, \bar{\boldsymbol{K}}_\phi \rangle \tag{17}$$

$$\le \frac{1}{\ell_1\ell_2} \sup_{\|\boldsymbol{J}\| \le 1} \langle \bar{\boldsymbol{L}}_\phi, \boldsymbol{J} \rangle = \frac{1}{\ell_1\ell_2}\|\bar{\boldsymbol{L}}_\phi\|_* \tag{18}$$

where the term $\|\mathcal{L}_\phi\|_{*,\boldsymbol{C}}$ represents the nuclear norm of the tensor $\mathcal{L}_\phi$ after being transformed through the linear operator $\boldsymbol{C}$ applied to its nonlinear transform product. The inequality in (17) arises from the fact that $\bar{\boldsymbol{K}}_\phi$ is a block diagonal matrix, while $\boldsymbol{J}$ is arbitrary. Considering the NLT transform TSVD $\mathcal{L}_\phi = \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{S}_\phi *_{\boldsymbol{C}} \mathcal{V}_\phi^{\mathrm{T}}$, the nuclear norm satisfies

$$\begin{aligned}\|\mathcal{L}_\phi\|_{*,\boldsymbol{C}} &= \sup_{\|\mathcal{K}_\phi\| \le 1} \langle \mathcal{L}_\phi, \mathcal{K}_\phi \rangle \\ &\ge \langle \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{S}_\phi *_{\boldsymbol{C}} \mathcal{V}_\phi^{\mathrm{T}}, \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{V}_\phi^{\mathrm{T}} \rangle \\ &= \langle \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{U}_\phi *_{\boldsymbol{C}} \mathcal{S}_\phi, \mathcal{V}_\phi^{\mathrm{T}} *_{\boldsymbol{C}} \mathcal{V}_\phi \rangle \\ &= \langle \mathcal{S}_\phi, \mathcal{I}_\phi \rangle = \frac{1}{\ell_1\ell_2}\operatorname{Tr}(\bar{\boldsymbol{S}}_\phi) = \frac{1}{\ell_1\ell_2}\|\bar{\boldsymbol{L}}_\phi\|_*. \end{aligned} \tag{19}$$

Combining (17) and (18) and (19), we then have the following definition of NLT transform TNN.

Definition 9 (NLT Transform TNN (NLTN)): Let $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ be a third-order tensor and $\mathcal{L}_\phi \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ its nonlinear transform under a predefined mapping. Given the linear transform matrix $\boldsymbol{C} \in \mathbb{R}^{I_3 \times I_3}$ in (8) applied along the third mode, the NLT transform TNN (NLTN) is defined as

$$\|\mathcal{L}\|_{\mathrm{NLTN}} = \|\mathcal{L}_\phi \times_3 \boldsymbol{C}\|_* = \|\mathcal{L}_\phi\|_{*,\boldsymbol{C}} = \frac{1}{\ell_1\ell_2}\sum_{i=1}^{I_3}\|\bar{\boldsymbol{L}}_\phi^{(i)}\|_*. \tag{20}$$

This norm captures the low-rank characteristics of $\mathcal{L}$ in its transformed nonlinear domain, leveraging the structural information encoded by $\boldsymbol{C}$ and $\mathcal{L}_\phi$.

Algorithm 1 presents the implementation of our proposed TNN method based on the NLT-domain transform, which reveals the internal operational structure of the model in an iterative optimization framework.

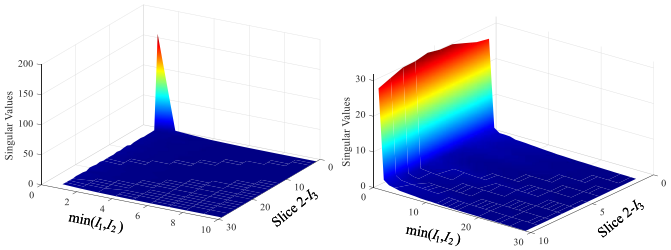
Algorithm 1 NLT Transform-Based TSVD (NTSVD) Within Optimization Framework

Input: Tensor $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$
Output: Tensor components $\mathcal{U}_\phi, \mathcal{S}_\phi, \mathcal{V}_\phi$
for *each iteration* $t = 1, 2, \ldots, T$ **do**
  1. Apply element-wise nonlinearity: $\mathcal{L}_\phi^{(t)} = \phi(\mathcal{L}^{(t)})$;
  2. Compute self-learning transform matrix: $[\boldsymbol{U}^{(t)}, \sim] = \mathrm{SVD}(\boldsymbol{L}_{(3)}^{(t)})$, and let $\boldsymbol{C}^{(t)} = \boldsymbol{F} \cdot \boldsymbol{U}^{(t)}$;
  3. Transform to NLT domain: $\bar{\mathcal{L}}^{(t)} = \mathcal{L}_\phi^{(t)} \times_3 \boldsymbol{C}^{(t)}$;
  4. For each frontal slice $\bar{\mathcal{L}}^{(i,t)}$, perform matrix SVD: $[\bar{\boldsymbol{U}}^{(i)}, \bar{\boldsymbol{S}}^{(i)}, \bar{\boldsymbol{V}}^{(i)}] = \mathrm{SVD}(\bar{\mathcal{L}}^{(i,t)})$;
  5. Aggregate slices to form tensors: $\bar{\mathcal{U}}_\phi, \bar{\mathcal{S}}_\phi, \bar{\mathcal{V}}_\phi$;
  6. Inverse transform to original domain:
    $\mathcal{U}_\phi^{(t)} = \bar{\mathcal{U}}_\phi \times_3 (\boldsymbol{C}^{(t)})^{-1}$
    $\mathcal{S}_\phi^{(t)} = \bar{\mathcal{S}}_\phi \times_3 (\boldsymbol{C}^{(t)})^{-1}$
    $\mathcal{V}_\phi^{(t)} = \bar{\mathcal{V}}_\phi \times_3 (\boldsymbol{C}^{(t)})^{-1}$;
  7. Update $\mathcal{L}^{(t+1)}$ based on optimization rule (e.g., ADMM step);
end

Remark 4: The transform-based TNN family is summarized in Fig. 1, showcasing its adaptability across different settings. Specifically: 1) if $\boldsymbol{C}$ is the identity matrix, the NLTN simplifies to NTNN [37], leveraging only nonlinear transform; 2) removing the elementwise nonlinear function reduces NLTN to cascade TNN [31], [38], showing its hierarchical relationship; and 3) when nonlinear transform property is removed, and $\boldsymbol{C}$ is fixed as DFT, DCT, or other unitary matrices, NLTN aligns with typical TNN variants such as DCT-based TNN [32], orthogonal TNN [33], or Frameleted-based TNN [35]. These relationships illustrate the flexibility and unifying nature of the proposed framework, accommodating multiple tensor representations under a single umbrella.

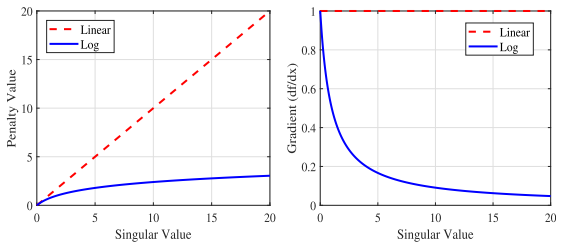
Similar to existing TNN models, NLTN also focuses on the low-rank property of the third mode, i.e., $\bar{\mathcal{L}}_\phi = \mathcal{L}_\phi \times_3 \boldsymbol{C}$. However, as observed from the singular value curves in Fig. 3, tensors with different permutation forms also exhibit low-rank properties. Therefore, instead of restricting the NLT transform to the third mode alone, we apply it to all three tensor modes to comprehensively explore the low-rank characteristics.

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(a) (b)

Fig. 3. Plot of singular values based on (a) permute-2 and (b) permute-3.



(a) (b)

Fig. 4. Comparison of linear and logarithmic penalties. (a) Penalty value curves. (b) Gradient curves.

Definition 10 (Full-scale NLTN (FNLTN)): Let $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ be a third-order tensor and $\mathcal{L}_\phi$ its nonlinear transform under a predefined mapping. The Full-scale NLTN (FNLTN) extends the NLTN by applying the learnable linear transform $\boldsymbol{C}_k \in \mathbb{R}^{I_k \times I_k}$ along all three tensor modes, which is defined as

$$\widetilde{\mathcal{L}_\phi} = \mathcal{L}_\phi \times_1 \boldsymbol{C}_1, \quad \vec{\mathcal{L}_\phi} = \mathcal{L}_\phi \times_2 \boldsymbol{C}_2, \quad \bar{\mathcal{L}_\phi} = \mathcal{L}_\phi \times_3 \boldsymbol{C}_3$$

$$\|\mathcal{L}\|_{\text{NLTN}_1} = \sum_{i=1}^{I_1} \left\| \widetilde{\mathcal{L}_\phi}(i,:,:) \right\|_* = \lambda_1 \sum_{i=1}^{I_1} \sum_j \sigma_j \left(\widetilde{\mathcal{L}_\phi}(i,:,:) \right)$$

$$\|\mathcal{L}\|_{\text{NLTN}_2} = \sum_{i=1}^{I_2} \left\| \vec{\mathcal{L}_\phi}(:,i,:) \right\|_* = \lambda_2 \sum_{i=1}^{I_2} \sum_j \sigma_j \left(\vec{\mathcal{L}_\phi}(:,i,:) \right)$$

$$\|\mathcal{L}\|_{\text{NLTN}_3} = \sum_{i=1}^{I_3} \left\| \bar{\mathcal{L}_\phi}(:,:,i) \right\|_* = \lambda_3 \sum_{i=1}^{I_3} \sum_j \sigma_j \left(\bar{\mathcal{L}_\phi}(:,:,i) \right)$$

$$\|\mathcal{L}\|_{\text{FNLTN}} = \lambda_1 \|\mathcal{L}\|_{\text{NLTN}_1} + \lambda_2 \|\mathcal{L}\|_{\text{NLTN}_2} + \lambda_3 \|\mathcal{L}\|_{\text{NLTN}_3} \quad (21)$$

where $\widetilde{\mathcal{L}_\phi}$, $\vec{\mathcal{L}_\phi}$, and $\bar{\mathcal{L}_\phi}$ denote the transformed tensors along first, second, and third modes, respectively. $\sigma_j(\boldsymbol{L})$ denote the jth singular value of $\boldsymbol{L}$. $\lambda_{1,2,3}$ are the tradeoff parameters.

The full-scale NLTN comprehensively explores low-rank properties across all tensor modes, offering greater flexibility and capturing richer structural information compared to single-mode transforms.

Building upon the full-scale NLTN defined in (21), we further propose a generalized formulation by replacing the nuclear norm with a logarithmic penalty, aiming to alleviate the over-shrinkage of dominant singular values and preserve essential structural features. This leads to the logarithmic-based GNLTN, defined as follows.

Definition 11 (Logarithmic-based Generalized NLTN): Given $\mathcal{L} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the Logarithmic-based Generalized NLTN (GNLTN) incorporates a logarithmic penalty to preserve key features while mitigating excessive penalization of large singular values. Formally, it is defined as

$$\begin{aligned}\|\mathcal{L}\|_{\text{GNLTN}} =& \left\|\mathcal{L}_\phi\right\|_{*,\boldsymbol{GC}_1} + \left\|\mathcal{L}_\phi\right\|_{*,\boldsymbol{GC}_2} + \left\|\mathcal{L}_\phi\right\|_{*,\boldsymbol{GC}_3} \\ =& \lambda_1 \sum_{i=1}^{I_1} \sum_j \log\left(\sigma_j \left(\widetilde{\mathcal{L}_\phi}(i,:,:) \right) + \epsilon \right) \\ &+ \lambda_2 \sum_{i=1}^{I_2} \sum_j \log\left(\sigma_j \left(\vec{\mathcal{L}_\phi}(:,i,:) \right) + \epsilon \right) \\ &+ \lambda_3 \sum_{i=1}^{I_3} \sum_j \log\left(\sigma_j \left(\bar{\mathcal{L}_\phi}(:,:,i) \right) + \epsilon \right). \quad (22)\end{aligned}$$

The term $\|\mathcal{L}_\phi\|_{*,\boldsymbol{GC}_i}$ denotes the nuclear norm of the nonlinear transform product $\mathcal{L}_\phi$ after being subjected to the linear operator $\boldsymbol{C}$ and guided by the logarithmic function Log. Here, ε is a very small positive number.

To better illustrate the mechanism by which the logarithmic penalty mitigates the over-penalization of large singular values, we compare it with the linear penalty both in terms of penalty value and gradient behavior. As shown in Fig. 4(a), the linear function $f(x) = x$ grows proportionally with the singular value, imposing equal penalization across all magnitudes. In contrast, the logarithmic function $f(x) = \log(1+(x/\tau))$ exhibits sublinear growth, resulting in relatively smaller penalization for larger singular values.

Fig. 4(b) further compares the gradients of these two functions. The linear penalty has a constant gradient of 1, meaning that all singular values are suppressed uniformly during optimization. However, the gradient of the logarithmic function, given by $f'(x) = (1/\tau + x)$, decreases with increasing x, which naturally leads to stronger suppression of small singular values while preserving the dominant ones. This adaptive suppression property helps to retain important structural information in the reconstruction process and avoid over-smoothing caused by excessive penalization of major components.

These observations highlight the advantage of adopting a logarithmic-like penalty in low-rank modeling tasks, where preserving large singular values is often critical for maintaining signal fidelity.

IV. GNLTN Minimization for HSISR

In this section, we introduce a novel HSISR solution based on the GNLTN framework. Specifically, the fusion problem is reformulated as a minimization task for subspace coefficient estimation. To effectively optimize the derived problem, we design a model solver based on the ADMM. The overall process is illustrated in Fig. 5.

A. GNLTN Model for HSISR

To derive the reduced image $\boldsymbol{A}_{(3)}$ in the observation model (6) and capture comprehensive image characteristics, we reformulate the estimation task under the proposed GNLTN framework. This process involves minimizing a hybrid energy functional, expressed as follows:

$$\begin{aligned}\underset{\boldsymbol{A}_{(3)}}{\text{argmin}} & \left\| \boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S} \right\|_F^2 + \left\| \boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)} \right\|_F^2 \\ &+ \lambda_1 \left\| \mathcal{A}_\phi \right\|_{*,\boldsymbol{GC}_1} + \lambda_2 \left\| \mathcal{A}_\phi \right\|_{*,\boldsymbol{GC}_2} + \lambda_3 \left\| \mathcal{A}_\phi \right\|_{*,\boldsymbol{GC}_3}. \quad (23)\end{aligned}$$

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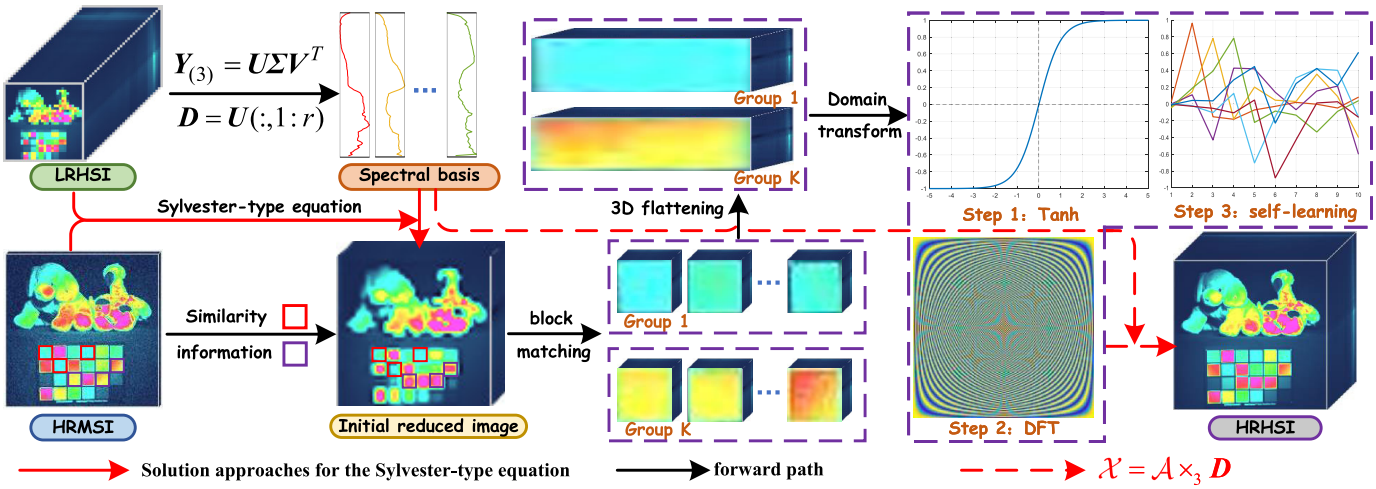


Fig. 5. Flowchart of the proposed GNLTN method for HSISR.

Instead of exclusively leveraging single-mode constraints, the GNLTN framework applies a tensor-norm penalty to promote a structured low-rank property in the reduced image and generalizes the regularization to exploit multidimensional correlations. This approach ensures robust modeling of both intrinsic spectral variations and spatial textures.

To enhance the estimation framework, we introduce a **nonlocal similarity prior** specifically designed for texture-rich images with complex structures. We consider the HRMSI tensor $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$, which is divided into overlapping spatial–spectral patches of size $(b)^{1/2} \times (b)^{1/2} \times s$. These patches undergo self-similarity analysis via an optimized clustering procedure

$$\left\{\mathcal{Z}^{(k)}\right\}_{k=1}^{K} = \text{K-means++}\left(\left\{\mathcal{Z}_i\right\}_{i=1}^{N_k}\right) \tag{24}$$

where each cluster $\mathcal{Z}^{(k)} = \{\mathcal{Z}^{(k,j)}\}_{j=1}^{N_k}$ consists of N_k similar patches, with K being the total number of clusters.

Given that the HRMSI and the reduced image $\boldsymbol{A}_{(3)}$ share spatial information, we extract low-dimensional abundance representations $\boldsymbol{A}_{(3)} \to \{\mathcal{A}^{(k,j)}\}_{j=1}^{N_k}$ from each cluster. These localized abundance patches are then organized into third-order tensors $\mathcal{A}^k \in \mathbb{R}^{b\times L\times N_k}, \quad \forall k \in \{1,\ldots,K\}$. This hierarchical tensor structure simultaneously captures spatial patterns, spectral signatures, and interpatch correlations. The reorganization of the data in this form facilitates a reformulated optimization framework that inherently preserves local spatial coherence while ensuring consistency in nonlocal patterns. The resulting tensor representation effectively decouples the multimodal relationships within the hyperspectral data, ensuring both mathematical tractability and the flexibility necessary for subsequent spectral unmixing tasks. Consequently, the optimization problem is reformulated as follows:

$$\begin{aligned}\underset{\boldsymbol{A}_{(3)}}{\operatorname{argmin}} &\left\|\boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\right\|_F^2 + \left\|\boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\right\|_F^2 \\ &+ \sum_{k=1}^{K} \lambda_1 \left\|\mathcal{A}_\phi^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_1} + \lambda_2 \left\|\mathcal{A}_\phi^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_2} + \lambda_3 \left\|\mathcal{A}_\phi^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_3}.\end{aligned} \tag{25}$$

B. ADMM-Based Solver

To effectively decouple and optimize the intertwined variables in Problem (25), the ADMM framework is utilized. This approach reformulates the problem into a sequence of iterative subproblems, each focusing on a specific subset of variables to facilitate efficient convergence. By introducing auxiliary variables $\mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3$ and corresponding Lagrangian multipliers $\mathcal{M}_i, \mathcal{O}_i$, the augmented objective function is expressed as

$$\begin{aligned}&L(\mathcal{A},\mathcal{U},\mathcal{V},\mathcal{W},\mathcal{P}_1,\mathcal{P}_2,\mathcal{P}_3,\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3,\mathcal{O}_1,\mathcal{O}_2,\mathcal{O}_3) \\ &= \left\|\boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\right\|_F^2 + \left\|\boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\right\|_F^2 \\ &\quad + \mu_1\left\|\mathcal{U} - \mathcal{A} + \frac{\mathcal{M}_1}{2\mu_1}\right\|_F^2 + \mu_1\left\|\mathcal{V} - \mathcal{A} + \frac{\mathcal{M}_2}{2\mu_1}\right\|_F^2 \\ &\quad + \mu_1\left\|\mathcal{W} - \mathcal{A} + \frac{\mathcal{M}_3}{2\mu_1}\right\|_F^2 \\ &\quad + \mu_2\left\|\mathcal{P}_1 - \mathcal{U}_\phi + \frac{\mathcal{O}_1}{2\mu_2}\right\|_F^2 + \lambda_1\sum_{k=1}^{K}\left\|\mathcal{P}_1^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_1} \\ &\quad + \mu_3\left\|\mathcal{P}_2 - \mathcal{V}_\phi + \frac{\mathcal{O}_2}{2\mu_3}\right\|_F^2 + \lambda_2\sum_{k=1}^{K}\left\|\mathcal{P}_2^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_2} \\ &\quad + \mu_4\left\|\mathcal{P}_3 - \mathcal{W}_\phi + \frac{\mathcal{O}_3}{2\mu_4}\right\|_F^2 + \lambda_3\sum_{k=1}^{K}\left\|\mathcal{P}_3^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_3}\end{aligned} \tag{26}$$

where μ_i controls the penalty associated with the auxiliary constraints. Optimization proceeds iteratively by alternating between subproblem updates for $\boldsymbol{A}_{(3)}$, auxiliary variables, and Lagrangian multipliers. The optimization is performed iteratively by solving subproblems for each variable in isolation while treating others as constants, where each step is summarized below:

1) Step 1: Optimization of $\boldsymbol{A}_{(3)}$: The subproblem for $\boldsymbol{A}_{(3)}$ focuses on minimizing the terms associated with

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reconstruction and coupling constraints

$$\begin{aligned}\boldsymbol{A}_{(3)} \in & \underset{\boldsymbol{A}_{(3)}}{\operatorname{argmin}} L(\mathcal{A}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3) \\ = & \underset{\boldsymbol{A}_{(3)}}{\operatorname{argmin}} \left\|\boldsymbol{X}_{(3)} - \boldsymbol{D}\boldsymbol{A}_{(3)}\boldsymbol{B}\boldsymbol{S}\right\|_F^2 + \left\|\boldsymbol{Y}_{(3)} - \boldsymbol{R}\boldsymbol{D}\boldsymbol{A}_{(3)}\right\|_F^2 \\ & + \mu_1 \left\|\boldsymbol{U}_{(3)} - \boldsymbol{A}_{(3)} + \frac{\boldsymbol{M}_{1(3)}}{2\mu_1}\right\|_F^2 \\ & + \mu_1 \left\|\boldsymbol{V}_{(3)} - \boldsymbol{A}_{(3)} + \frac{\boldsymbol{M}_{2(3)}}{2\mu_1}\right\|_F^2 \\ & + \mu_1 \left\|\boldsymbol{W}_{(3)} - \boldsymbol{A}_{(3)} + \frac{\boldsymbol{M}_{3(3)}}{2\mu_1}\right\|_F^2. \end{aligned} \tag{27}$$

By differentiating with respect to $\boldsymbol{A}_{(3)}$, the optimal solution is derived as a closed-form matrix equation

$$\boldsymbol{H}_1\boldsymbol{A}_{(3)} + \boldsymbol{A}_{(3)}\boldsymbol{H}_2 = \boldsymbol{H}_3 \tag{28}$$

where

$$\begin{aligned}\boldsymbol{H}_1 &= \left((\boldsymbol{R}\boldsymbol{D})^{\mathrm{T}}(\boldsymbol{R}\boldsymbol{D}) + \mu\boldsymbol{I}_L\right) \\ \boldsymbol{H}_2 &= (\boldsymbol{B}\boldsymbol{S})(\boldsymbol{B}\boldsymbol{S})^{\mathrm{T}} \\ \boldsymbol{H}_3 &= (\boldsymbol{R}\boldsymbol{D})^{\mathrm{T}}\boldsymbol{Y}_{(3)} + \boldsymbol{D}^{\mathrm{T}}\boldsymbol{X}_{(3)}(\boldsymbol{B}\boldsymbol{S})^{\mathrm{T}} + \mu_1\left(\boldsymbol{U}_{(3)} + \frac{\boldsymbol{M}_{1(3)}}{2\mu_1}\right. \\ &\quad \left. + \boldsymbol{V}_{(3)} + \frac{\boldsymbol{M}_{2(3)}}{2\mu_1} + \boldsymbol{W}_{(3)} + \frac{\boldsymbol{M}_{3(3)}}{2\mu_1}\right).\end{aligned}$$

The equation is solved using closed-form techniques as described in [27] and [44].

2) Step 2: Optimization of Auxiliary Variables- ($\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3$): The optimization of auxiliary variables $\mathcal{P}_i$ is divided into independent subproblems for each cluster. Taking $\mathcal{P}_1$ as an example, the optimization is formulated as

$$\begin{aligned}\mathcal{P}_1 \in & \underset{\mathcal{P}_1}{\operatorname{argmin}} L(\mathcal{P}_1, \mathcal{U}, \mathcal{M}_1, \mathcal{O}_1) \\ = & \mu_2 \left\|\mathcal{P}_1 - \mathcal{U}_\phi + \frac{\mathcal{O}_1}{2\mu_2}\right\|_F^2 + \lambda_1 \sum_{k=1}^{K} \left\|\mathcal{P}_1^k\right\|_{*,\boldsymbol{G}\boldsymbol{C}_1}. \end{aligned} \tag{29}$$

Each cluster $\mathcal{P}_1^k$ is then optimized individually. After transforming the tensors into a new basis, the subproblem for cluster k is

$$\begin{aligned}\underset{\widetilde{\mathcal{P}}_1^k}{\text{minimize}} \quad & \frac{\mu_2}{N_k}\left\|\widetilde{\mathcal{P}}_1^k - \mathcal{H}_1^k\right\|_F^2 \\ & + \lambda_1 \sum_{i=1}^{N_k}\sum_{i=1}^{I_1}\sum_{j} \log\left(\sigma_j\left(\widetilde{\mathcal{P}}_1(i,:,:)\right) + \epsilon\right) \end{aligned} \tag{30}$$

where $\widetilde{\mathcal{P}}_1^k = \mathcal{P}_1^k \times_1 \boldsymbol{C}_1$ and $\mathcal{H}_1^k = (\mathcal{U}_\phi^k - (\mathcal{O}_1^k/2\mu_2)) \times_1 \boldsymbol{C}_1$ are transformed tensors. Problem (30) is equivalent to optimizing the TNN under a nonlinear logarithmic transform, which is efficiently solved using Theorem 2 (log-sum tensor singular value thresholding, LTSVT) from [38].

After solving for $\widetilde{\mathcal{P}}_1^k$, the inverse transform is applied to reconstruct the cluster contribution: $\mathcal{P}_1^k = \widetilde{\mathcal{P}}_1^k \times_1 \boldsymbol{C}_1^{-1}$. The final solution $\mathcal{P}_1$ is obtained by aggregating all cluster components $\mathcal{P}_1 = \bigcup_{k=1}^{K} \mathcal{P}_1^k$. The optimization of $\mathcal{P}_2$ and $\mathcal{P}_3$ follows the same procedure as described for $\mathcal{P}_1$.

3) Step 3: Optimization of Auxiliary Variables- ($\mathcal{U}, \mathcal{V}, \mathcal{W}$): To refine the auxiliary variables $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ in problem (26), we first extract all terms involving $\mathcal{U}$, resulting in the following optimization subproblem

$$\underset{\mathcal{U}}{\operatorname{argmin}} \mu_1 \left\|\mathcal{U} - \mathcal{A} + \frac{\mathcal{M}_1}{2\mu_1}\right\|_F^2 + \mu_2 \left\|\mathcal{P}_1 - \mathcal{U}_\phi + \frac{\mathcal{O}_1}{2\mu_2}\right\|_F^2. \tag{31}$$

Let the (i, j)th element of the mode-3 unfolding of tensors $\mathcal{U}, \mathcal{G} = \mathcal{A} - (\mathcal{M}_1/2\mu_1)$, and $\mathcal{Q} = \mathcal{P}_1 + (\mathcal{O}_1/2\mu_2)$ be denoted as u_{ij}, g_{ij}, and k_{ij}, respectively. Then, Problem (31) is decomposed into $n_1 n_2 r$ independent scalar minimization problems

$$\underset{u_{\mathrm{ij}}}{\operatorname{argmin}} \quad \mu_1\left(u_{\mathrm{ij}} - g_{\mathrm{ij}}\right)^2 + \mu_2\left(N_1\left(u_{\mathrm{ij}}\right) - q_{\mathrm{ij}}\right)^2 \tag{32}$$

where $N_1(\cdot)$ represents a nonlinear function. Given that the estimation of $\mathcal{G}$ might be imprecise, it is unnecessary to compute the exact solution to the $\mathcal{U}$ subproblem at every iteration. Instead, a few Newton iterations are sufficient, as for better computational efficiency [37]. Similarly, the subproblems for $\mathcal{V}$ and $\mathcal{W}$ can be solved using the same approach, leveraging their analogous structures. This optimization step ensures the update of $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ in a computationally efficient manner while balancing accuracy and practicality.

4) Step 4: Lagrangian Multiplier Updates: The Lagrangian multipliers are updated to enforce consistency constraints across variables and to ensure convergence within the augmented Lagrangian framework. The scaled multipliers are updated iteratively as follows:

$$\begin{aligned}&\mathcal{M}_1 = \mathcal{M}_1 + 2\mu_1(\mathcal{U} - \mathcal{A}), \quad \mathcal{M}_2 = \mathcal{M}_2 + 2\mu_1(\mathcal{V} - \mathcal{A}) \\ &\mathcal{M}_3 = \mathcal{M}_3 + 2\mu_1(\mathcal{W} - \mathcal{A}), \quad \mathcal{O}_1 = \mathcal{O}_1 + 2\mu_2\left(\mathcal{P}_1 - \mathcal{U}_\phi\right) \\ &\mathcal{O}_2 = \mathcal{O}_2 + 2\mu_3\left(\mathcal{P}_2 - \mathcal{V}_\phi\right), \quad \mathcal{O}_3 = \mathcal{O}_3 + 2\mu_4\left(\mathcal{P}_3 - \mathcal{W}_\phi\right).\end{aligned} \tag{33}$$

Each multiplier update reflects the discrepancy between the current variable estimates and their expected values, driving the optimization toward a feasible solution.

Summarizing the aforementioned descriptions, we present the entire ADMM-based solution process for solving problem (30) in Algorithm 2.

Algorithm 2 ADMM for Solving (26)

Input: LR-HSI $\mathcal{Y}$; HR-MSI $\mathcal{Z}$; the parameters μ_1, μ_2, μ_3, μ_4; Cluster number K; path size p; step size s, Max iteration number T.

Output: The target HR-HSI $\mathcal{X}$.

Initialization: $\mu_1 = 1e-2$, $\mu_2 = 1e-2$, $\mu_3 = 1e-3$, $\mu_4 = 1e-3$, $K = 400$, $p = 6 or\ 8$, $s = 4$, $T = 50$;

while $t \leqslant T$ **do**

  Update $\boldsymbol{A}_{(3)}$ via (28);

  Update $\mathcal{P}$ via LTSVT [38];

  Update $\mathcal{U}, \mathcal{V}, \mathcal{W}$ via Newton method [37];

  Update $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$ via (33);

  $t \leftarrow t + 1$;

end

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C. Computational Complexity

To enable a tractable analysis, we consider an arbitrary clustered group $\mathcal{A}^k \in \mathbb{R}^{b\times L\times N_k}$ and estimate the per-iteration computational complexity based on its internal structure. Updating the mode-3 unfolding $\boldsymbol{A}_{(3)}$ has a computational cost of $\mathcal{O}(\mathrm{HW}\max\log(\mathrm{HW}), L^2)$ per iteration. The updates of $\mathcal{U}, \mathcal{V}$, and $\mathcal{W}$ are decomposed into bLN_k independent scalar subproblems, each solved via I iterations of Newton's method. Each iteration requires approximately $(27I+4)$ floating-point operations, resulting in a total cost of $(81I+12), \mathrm{bLN}_k$ FLOPs, i.e., an overall complexity of $\mathcal{O}(I, \mathrm{bLN}_k)$ per iteration. The optimization of $\mathcal{P}_1, \mathcal{P}_2,$, and $\mathcal{P}_3$ involves mode-specific operations, including nonlinear element-wise activations (e.g., tanh), DFTs, self-learned transformations, and singular value decompositions (SVDs). For $\mathcal{P}_1 \in \mathbb{R}^{b\times L\times N_k}$, the per-iteration cost is $\mathcal{O}(\mathrm{bLN}_k(\log N_k + b + L) + N_k b^2 L)$, where the first term accounts for domain transforms and the second term corresponds to SVDs on frontal slices of size $b \times L$. Similarly, the complexities for $\mathcal{P}_2$ and $\mathcal{P}_3$ are $\mathcal{O}(\mathrm{bLN}_k(\log N_k + L + N_k) + L^2\mathrm{bN}_k)$ and $\mathcal{O}(\mathrm{bLN}_k(\log N_k + N_k + L) + \mathrm{bLN}_k^2)$, respectively. Lagrangian multipliers are updated via simple element-wise operations such as tensor subtraction and scalar multiplication, with a total per-iteration cost of $\mathcal{O}(\mathrm{bLN}_k)$.

V. Experiments

To rigorously assess the capabilities of GNLTN, a comparative analysis was conducted using several advanced methodologies as benchmarks. The selected methods include coupled sparse tensor factorization (CSTF)[1] [28], low tensor multirank regularization (LTMR)[2] [27], coupled tensor ring factorization (CTRF)[3] [45], coupled tensor double-factor decomposition (CTDF)[4] [2], log-sum-based full-scale cascade-transform-based TNN (LFCTNN-FC)[5] [38], LFCTNN-FL [38], and generalized TNN (GTNN)[6] [1]. To ensure fair and objective comparisons, all unspecified parameters for these methods were optimized for their best performance. Performance was measured using six established metrics: peak signal to noise ratio (PSNR) [24], structural similarity (SSIM) [46], spectral angle mapper (SAM) [1], erreur relative globale admensionnelle de synthese (ERGSA) [28], the degree of distortion (DD), and cross correlation (CC) [42]. While higher PSNR, SSIM, and CC values indicate superior results, lower SAM, ERGAS, and DD scores correspond to better performance. The experiments were conducted using MATLAB R2020a on a Windows 10 system with an AMD R7-5800 processor and 16 GB of memory.

A. Dataset

The evaluation process incorporated both simulated and real-world datasets to ensure a comprehensive analysis. For the simulated experiments, three widely recognized datasets were selected: Columbia Computer Vision Laboratory (CAVE) [47], Pavia University (PU) [48], Pavia city center (PC) dataset[7], and Cuprite mine (CM) dataset. The detailed information of the dataset, including the sampling method and noise conditions, is provided in Table I.

TABLE I
Overview of Four Datasets for Image Processing

Data	PU	PC	CM	CAVE
Number of Data	1	1	1	32
Orig. Spatial Size	610×340	1096×715	256×256	512×512
Exp. Spatial Size	256×256	256×256	256×256	512×512
Orig. Spectral Size	115	191	224	31
Exp. Spectral Size	93	103	89	31
Blur Method	Gaussian and Uniform			
Kernel Size	7×7	7×7	7×7	7×7
Std. Deviation	2	2	2	2
SNR (LRHSI)	20-35	20-35	20-35	20
SNR (HRMSI)	20-35	20-35	20-35	20
Scale Ratio	4	4	4	16
Resp. Filter	IKONOS	IKONOS	IKONOS	Nikon D700

Real-world HSISR experiments utilized Sentinel-2 MSI and Hyperion HSI from the Earth Observing-1 satellite [3]. The Hyperion data originally includes 220 spectral bands spanning 400–2500 nm with a spatial resolution of 30 m. After excluding noisy bands, 89 bands were retained for analysis. Sentinel-2 MSI, which consists of 13 spectral bands, provided four bands at 490, 560, 665, and 842 nm, each with a 10-m spatial resolution, for the super-resolution process. A 180 × 180 region from the Hyperion HSI and a corresponding 540 × 540 area from Sentinel-2 MSI were extracted for evaluation. Blur kernels and spectral response functions were estimated using the approach detailed in [49].

B. Synthetic Data Experiments

1) Quantitative Analysis: This section presents a detailed quantitative comparison of various methods evaluated on four key hyperspectral image datasets: PU, PC, CM, and the 32 CAVE images. The methods are assessed under varying noise levels (SNRs) and different sampling strategies. Specifically, the evaluation includes Gaussian sampling with SNR = 30 dB for LRHSI and SNR = 35 dB for HRMSI. In addition, uniform sampling is evaluated under three different noise levels: SNR = 35 dB, SNR = 30 dB, and SNR = 20 dB for both LRHSI and HRMSI. Tables II and III include performance metrics such as PSNR, SSIM, SAM, ERGAS, DD, and CC for each method, with optimal values marked in red and second-best values in blue.

a) Gaussian sampling: As shown in Tables II, for the CAVE dataset, the proposed GNLTN method achieves superior performance across all evaluated metrics, with a notable PSNR score of 36.0563. This marks an improvement of 1.73% over the next best-performing method, LFCTNN-FL. Such an increase underscores GNLTN's enhanced capability in maintaining image fidelity. In addition, the method shows a marked reduction in the ERGAS metric, achieving a value of

[1]https://github.com/renweidian/CSTF
[2]https://github.com/renweidian/LTMR
[3]https://prowdiy.github.io/weihe.github.io/publication.html
[4]https://github.com/tingxu113/CTDF
[5]https://github.com/xuhonghui96/LFCTNN
[6]https://github.com/renweidian/GTNN
[7]http://www.ehu.eus/ccwintco/index.php

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TABLE II

QUANTITATIVE METRICS (PSNR, SSIM, SAM, ERGAS, DD, AND CC) OF ALL COMPARED METHODS. THE BEST VALUES AND THE SECOND BEST VALUES ARE, RESPECTIVELY, HIGHLIGHTED BY RED AND BLUE COLORS

Image	Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	DD↓	CC↑
CAVE	$\text{CSTF}_{\text{TCYB22}}$	$33.2476_{\downarrow 7.79\%}$	$0.7563_{\downarrow 13.23\%}$	$22.1949_{\uparrow 12.90\%}$	$1.5932_{\uparrow 31.67\%}$	$4.5818_{\uparrow 42.12\%}$	$0.9798_{\downarrow 0.86\%}$
	$\text{LTMR}_{\text{TIP22}}$	$33.7787_{\downarrow 6.32\%}$	$0.7557_{\downarrow 13.30\%}$	$20.4871_{\uparrow 4.21\%}$	$1.5324_{\uparrow 26.64\%}$	$4.3532_{\uparrow 35.03\%}$	$0.9841_{\downarrow 0.42\%}$
	$\text{CTRF}_{\text{PR22}}$	$34.6299_{\downarrow 3.96\%}$	$0.8259_{\downarrow 5.25\%}$	$20.8123_{\uparrow 5.87\%}$	$1.4830_{\uparrow 22.56\%}$	$4.1655_{\uparrow 29.21\%}$	$0.9775_{\downarrow 1.09\%}$
	$\text{CTDF}_{\text{TGRS24}}$	$34.9364_{\downarrow 5.01\%}$	$0.8365_{\downarrow 4.03\%}$	$19.6587_{\uparrow 0.00\%}$	$1.4225_{\uparrow 19.00\%}$	$3.9058_{\uparrow 30.71\%}$	$0.9859_{\downarrow 0.55\%}$
	$\text{LFCTNN-FC}_{\text{TGRS24}}$	$35.0560_{\downarrow 2.77\%}$	$0.8394_{\downarrow 3.69\%}$	$25.5789_{\uparrow 30.11\%}$	$1.4120_{\uparrow 16.69\%}$	$3.8515_{\uparrow 19.47\%}$	$0.9848_{\downarrow 0.35\%}$
	$\text{LFCTNN-FL}_{\text{TGRS24}}$	$35.4321_{\downarrow 1.73\%}$	$0.8705_{\downarrow 0.13\%}$	$24.6445_{\uparrow 25.36\%}$	$1.3838_{\uparrow 14.36\%}$	$3.6982_{\uparrow 14.71\%}$	$0.9868_{\downarrow 0.15\%}$
	$\text{GTNN}_{\text{TNNLS25}}$	$33.9794_{\downarrow 5.66\%}$	$0.7691_{\downarrow 11.77\%}$	$20.0672_{\uparrow 2.08\%}$	$1.4727_{\uparrow 21.71\%}$	$4.2454_{\uparrow 31.69\%}$	$0.9843_{\downarrow 0.40\%}$
	GNLTN	$36.0563_{\downarrow 0.00\%}$	$0.8716_{\downarrow 0.00\%}$	$20.7843_{\uparrow 5.73\%}$	$1.2100_{\uparrow 0.00\%}$	$3.2239_{\uparrow 0.00\%}$	$0.9883_{\downarrow 0.00\%}$
Pavia University	$\text{CSTF}_{\text{TCYB22}}$	$41.9193_{\downarrow 3.92\%}$	$0.9809_{\downarrow 0.67\%}$	$2.4010_{\uparrow 29.44\%}$	$1.2369_{\uparrow 23.22\%}$	$1.5241_{\uparrow 27.82\%}$	$0.9962_{\downarrow 0.12\%}$
	$\text{LTMR}_{\text{TIP22}}$	$42.1415_{\downarrow 3.42\%}$	$0.9803_{\downarrow 0.73\%}$	$2.3780_{\uparrow 28.20\%}$	$1.9270_{\uparrow 91.97\%}$	$1.5003_{\uparrow 25.82\%}$	$0.9972_{\downarrow 0.02\%}$
	$\text{CTRF}_{\text{RP22}}$	$41.9957_{\downarrow 3.75\%}$	$0.9813_{\downarrow 0.63\%}$	$2.3309_{\uparrow 25.66\%}$	$1.2344_{\uparrow 22.97\%}$	$1.4925_{\uparrow 25.17\%}$	$0.9961_{\downarrow 0.13\%}$
	$\text{CTDF}_{\text{TGRS24}}$	$42.8970_{\downarrow 1.68\%}$	$0.9846_{\downarrow 0.11\%}$	$2.0848_{\uparrow 12.39\%}$	$1.1081_{\uparrow 10.39\%}$	$1.3793_{\uparrow 15.67\%}$	$0.9969_{\downarrow 0.07\%}$
	$\text{LFCTNN-FC}_{\text{TGRS24}}$	$43.3575_{\downarrow 0.63\%}$	$0.9866_{\downarrow 0.09\%}$	$1.9494_{\uparrow 5.09\%}$	$1.0419_{\uparrow 3.80\%}$	$1.2413_{\uparrow 4.10\%}$	$0.9973_{\downarrow 0.01\%}$
	$\text{LFCTNN-FL}_{\text{TGRS24}}$	$43.4734_{\downarrow 0.36\%}$	$0.9869_{\downarrow 0.06\%}$	$1.9229_{\uparrow 3.67\%}$	$1.0233_{\uparrow 1.94\%}$	$1.2240_{\uparrow 2.65\%}$	$0.9973_{\downarrow 0.01\%}$
	$\text{GTNN}_{\text{TNNLS25}}$	$42.9282_{\downarrow 1.61\%}$	$0.9846_{\downarrow 0.29\%}$	$2.0943_{\uparrow 12.91\%}$	$1.0892_{\uparrow 8.51\%}$	$1.3309_{\uparrow 11.62\%}$	$0.9970_{\downarrow 0.04\%}$
	GNLTN	$43.6317_{\downarrow 0.00\%}$	$0.9875_{\downarrow 0.00\%}$	$1.8549_{\uparrow 0.00\%}$	$1.0038_{\uparrow 0.00\%}$	$1.1924_{\uparrow 0.00\%}$	$0.9974_{\downarrow 0.00\%}$
Pavia city center	$\text{CSTF}_{\text{TCYB22}}$	$41.0624_{\downarrow 7.96\%}$	$0.9865_{\downarrow 0.74\%}$	$4.1538_{\uparrow 69.97\%}$	$1.0359_{\uparrow 21.93\%}$	$1.2032_{\uparrow 6.99\%}$	$0.9977_{\downarrow 0.03\%}$
	$\text{LTMR}_{\text{TIP22}}$	$40.9137_{\downarrow 8.30\%}$	$0.9862_{\downarrow 0.77\%}$	$4.0497_{\uparrow 65.71\%}$	$1.3399_{\uparrow 57.71\%}$	$1.7641_{\uparrow 56.86\%}$	$0.9950_{\downarrow 0.30\%}$
	$\text{CTRF}_{\text{RP22}}$	$43.5751_{\downarrow 2.33\%}$	$0.9921_{\downarrow -0.18\%}$	$2.9395_{\uparrow 20.28\%}$	$0.9464_{\uparrow 11.39\%}$	$1.2992_{\uparrow 15.53\%}$	$0.9975_{\downarrow 0.05\%}$
	$\text{CTDF}_{\text{TGRS24}}$	$43.3643_{\downarrow 2.80\%}$	$0.9914_{\downarrow 0.25\%}$	$3.2416_{\uparrow 32.18\%}$	$0.9776_{\uparrow 15.02\%}$	$1.4368_{\uparrow 27.76\%}$	$0.9973_{\downarrow 0.05\%}$
	$\text{LFCTNN-FC}_{\text{TGRS24}}$	$44.1247_{\downarrow 1.10\%}$	$0.9930_{\downarrow 0.09\%}$	$2.6544_{\uparrow 8.62\%}$	$0.9110_{\uparrow 7.23\%}$	$1.2032_{\uparrow 6.99\%}$	$0.9977_{\downarrow 0.03\%}$
	$\text{LFCTNN-FL}_{\text{TGRS24}}$	$44.0918_{\downarrow 1.17\%}$	$0.9931_{\downarrow -0.08\%}$	$2.6398_{\uparrow 8.02\%}$	$0.9060_{\uparrow 6.64\%}$	$1.2009_{\uparrow 6.78\%}$	$0.9954_{\downarrow 0.26\%}$
	$\text{GTNN}_{\text{TNNLS25}}$	$43.5709_{\downarrow 2.34\%}$	$0.9921_{\downarrow 0.18\%}$	$2.8905_{\uparrow 18.28\%}$	$0.9624_{\uparrow 13.28\%}$	$1.2871_{\uparrow 14.45\%}$	$0.9974_{\downarrow 0.06\%}$
	GNLTN	$44.6151_{\downarrow 0.00\%}$	$0.9939_{\downarrow 0.00\%}$	$2.4438_{\uparrow 0.00\%}$	$0.8496_{\uparrow 0.00\%}$	$1.1246_{\uparrow 0.00\%}$	$0.9980_{\downarrow 0.00\%}$
Cuprite mine	$\text{CSTF}_{\text{TCYB22}}$	$37.2748_{\downarrow 4.08\%}$	$0.9629_{\downarrow 1.17\%}$	$1.9454_{\uparrow 23.46\%}$	$0.9758_{\uparrow 21.30\%}$	$2.8459_{\uparrow 21.61\%}$	$0.9896_{\downarrow 0.33\%}$
	$\text{LTMR}_{\text{TIP22}}$	$36.2532_{\downarrow 6.71\%}$	$0.9524_{\downarrow 2.25\%}$	$2.2541_{\uparrow 43.09\%}$	$1.1277_{\uparrow 40.20\%}$	$3.2351_{\uparrow 38.22\%}$	$0.9864_{\downarrow 0.65\%}$
	$\text{CTRF}_{\text{PR22}}$	$37.2917_{\downarrow 4.04\%}$	$0.9607_{\downarrow 1.39\%}$	$1.9972_{\uparrow 26.75\%}$	$1.0158_{\uparrow 26.28\%}$	$2.8855_{\uparrow 23.30\%}$	$0.9891_{\downarrow 0.38\%}$
	$\text{CTDF}_{\text{TGRS24}}$	$37.4841_{\downarrow 3.54\%}$	$0.9649_{\downarrow 0.97\%}$	$1.8656_{\uparrow 18.39\%}$	$0.9228_{\uparrow 14.73\%}$	$2.7868_{\uparrow 19.06\%}$	$0.9904_{\downarrow 0.25\%}$
	$\text{LFCTNN-FC}_{\text{TGRS24}}$	$38.0708_{\downarrow 2.03\%}$	$0.9693_{\downarrow 0.52\%}$	$1.7568_{\uparrow 11.51\%}$	$0.8812_{\uparrow 9.54\%}$	$2.5762_{\uparrow 10.06\%}$	$0.9915_{\downarrow 0.14\%}$
	$\text{LFCTNN-FL}_{\text{TGRS24}}$	$38.0900_{\downarrow 1.98\%}$	$0.9689_{\downarrow 0.56\%}$	$1.7563_{\uparrow 11.48\%}$	$0.8855_{\uparrow 10.08\%}$	$2.5756_{\uparrow 10.04\%}$	$0.9915_{\downarrow 0.14\%}$
	$\text{GTNN}_{\text{TNNLS25}}$	$37.6537_{\downarrow 3.11\%}$	$0.9657_{\downarrow 0.89\%}$	$1.8632_{\uparrow 18.27\%}$	$0.9350_{\uparrow 16.23\%}$	$2.7200_{\uparrow 16.23\%}$	$0.9905_{\downarrow 0.24\%}$
	GNLTN	$38.8597_{\downarrow 0.00\%}$	$0.9743_{\downarrow 0.00\%}$	$1.5756_{\uparrow 0.00\%}$	$0.8044_{\uparrow 0.00\%}$	$2.3405_{\uparrow 0.00\%}$	$0.9929_{\downarrow 0.00\%}$

1.2100—an improvement of 21.71% over GTNN and 14.36% over LFCTNN-FL. This suggests that GNLTN not only preserves high image quality but also delivers more precise and computationally efficient reconstructions.

On the PU dataset, GNLTN stands out as the top performer, surpassing all other methods in key metrics. Notably, GNLTN achieves a PSNR of 43.6317 dB, outperforming CSTF by 7.96% and GTNN by 1.61%. This performance is particularly notable in the SAM metric, where GNLTN leads by significant margins across various methods, including CSTF (29.44%), LTMR (28.20%), CTRF (32.18%), CTDF (2.56%), LFCTNN-FC (5.09%), LFCTNN-FL (3.67%), and GTNN (12.91%). These improvements indicate that GNLTN's architecture is more effective at capturing tensor structures, which translates to more accurate image reconstructions.

On the PC dataset, GNLTN sets a new benchmark by securing the highest performance across all evaluation metrics. Its impressive results in SAM and ERGAS metrics establish a commanding lead over other models, further affirming its capability to tackle complex hyperspectral image reconstruction tasks with superior quality and efficiency. Collectively, GNLTN's exceptional performance across these three datasets solidifies its position as the leading method for hyperspectral image processing, capable of advancing the state of the art in this domain.

To further substantiate the quantitative effectiveness of our proposed method, Fig. 6 presents band-wise PSNR results on representative datasets, including PC and thread spools. As illustrated, GNLTN consistently outperforms competing methods across nearly all spectral bands. These results, in conjunction with the preceding spatial and spectral analyses, provide strong evidence for the superiority of our approach.

b) Uniform sampling: As demonstrated in Table III, the GNLTN method shows superior performance across all metrics on PU, achieving a PSNR of 37.4822 dB at SNR = 20 dB. This represents a 0.16 dB improvement over the second-best method, LFCTNN-FL, which achieves 36.0278 dB. Furthermore, GNLTN outperforms other methods in ERGAS, reducing the value to 2.3010, which corresponds to a 18.87% improvement compared to LFCTNN-FL. Turning to the PC

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TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON PAVIA UNIVERSITY, PAVIA CITY CENTER, AND CUPRITE MINE DATASETS UNDER VARYING NOISE LEVELS WITH UNIFORM SAMPLING

Method	Pavia University					Pavia city center					Cuprite mine				
	PSNR↑	SSIM↑	SAM↓	ERGAS↓	DD↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	DD↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	DD↓
SNR = 35dB for LRHSI **SNR = 35dB** for HRMSI															
$\text{CSTF}_{\text{TCYB22}}$	34.9486	0.9397	4.8467	2.9067	3.4798	30.6809	0.8863	12.2956	4.5453	6.5535	33.2412	0.9268	2.7637	1.5393	4.5818
$\text{LTMR}_{\text{TIP22}}$	41.4691	0.9793	2.5468	1.3150	1.6089	40.9331	0.9867	3.9584	1.3558	1.7467	37.1168	0.9607	1.9904	1.0244	2.8917
$\text{CTRF}_{\text{PR22}}$	36.3831	0.9553	4.1549	2.4853	2.9828	35.8525	0.9655	7.4767	2.4794	3.5485	34.9490	0.9513	2.2876	1.2791	3.7094
$\text{CTDF}_{\text{TGRS24}}$	43.0653	0.9852	2.0123	1.0879	1.3401	43.0038	0.9916	3.1870	1.0116	1.4706	37.7490	0.9655	1.8229	0.9152	2.7260
$\text{LFCTNN-FC}_{\text{TGRS24}}$	43.6333	0.9875	1.8625	1.0187	1.1866	44.2613	0.9932	2.6082	0.8954	1.1820	38.6422	0.9721	1.6304	0.8313	2.4042
$\text{LFCTNN-FL}_{\text{TGRS24}}$	43.7623	0.9878	1.8359	0.9940	1.1676	44.4790	0.9935	2.5197	0.8751	1.1457	38.7305	0.9724	1.6118	0.8267	2.3817
$\text{GTNN}_{\text{TNNLS25}}$	43.2378	0.9859	1.9916	1.0544	1.2687	43.9629	0.9926	2.7365	0.9386	1.2220	38.2978	0.9696	1.7084	0.8764	2.5137
GNLTN	43.9738	0.9887	1.7613	0.9683	1.1283	44.9442	0.9943	2.3278	0.8169	1.0786	39.2995	0.9760	1.4879	0.7680	2.2189
SNR = 30dB for LRHSI **SNR = 30dB** for HRMSI															
$\text{CSTF}_{\text{TCYB22}}$	33.9976	0.9160	5.5772	3.2325	3.9822	29.8627	0.8570	13.6636	5.1240	7.4217	31.7503	0.8992	3.2463	1.7461	5.4144
$\text{LTMR}_{\text{TIP22}}$	39.4128	0.9650	3.1620	1.6092	2.0840	39.1632	0.9788	4.7454	1.6094	2.1688	34.0086	0.9266	2.6598	1.3255	4.0834
$\text{CTRF}_{\text{PR22}}$	35.6633	0.9410	4.5629	2.6229	3.2945	35.1556	0.9574	7.9616	2.6356	3.8235	32.7084	0.9168	2.9047	1.5415	4.7698
$\text{CTDF}_{\text{TGRS24}}$	40.4551	0.9701	2.7025	1.4599	1.8758	40.2143	0.9796	4.8861	1.4560	2.1509	34.2147	0.9274	2.5491	1.2634	4.0340
$\text{LFCTNN-FC}_{\text{TGRS24}}$	41.2591	0.9759	2.4090	1.3066	1.6480	41.4815	0.9865	3.5300	1.2098	1.6791	35.0266	0.9409	2.2784	1.1414	3.6071
$\text{LFCTNN-FL}_{\text{TGRS24}}$	41.3307	0.9762	2.3925	1.2946	1.6276	41.5480	0.9869	3.5098	1.1965	1.6595	35.0882	0.9415	2.2659	1.1382	3.5849
$\text{GTNN}_{\text{TNNLS25}}$	40.7586	0.9729	2.6053	1.3811	1.7616	40.9642	0.9849	3.8230	1.2893	1.7833	34.8514	0.9385	2.3488	1.1793	3.6921
GNLTN	41.7445	0.9785	2.2158	1.2286	1.5417	42.2872	0.9889	3.0713	1.0968	1.5177	35.3218	0.9440	2.1846	1.1101	3.4953
SNR = 20dB for LRHSI **SNR = 20dB** for HRMSI															
$\text{CSTF}_{\text{TCYB22}}$	29.3969	0.7573	9.7224	5.0776	6.9451	24.3442	0.6492	22.2223	9.9353	14.3824	23.2494	0.5779	9.0116	4.3364	14.1514
$\text{LTMR}_{\text{TIP22}}$	30.9247	0.8078	8.4940	4.3577	5.7967	30.9511	0.8803	10.8495	4.0124	5.7561	25.3972	0.6599	7.0632	3.4981	10.9888
$\text{CTRF}_{\text{PR22}}$	30.2662	0.7813	8.6747	4.6670	6.2832	29.2259	0.8388	13.4756	4.9582	7.3465	23.8246	0.6005	8.3952	4.1832	13.3673
$\text{CTDF}_{\text{TGRS24}}$	32.0477	0.8230	6.9748	3.8229	5.0950	30.6747	0.8700	11.9060	4.3740	6.1613	25.6516	0.6775	6.5330	3.2674	10.8050
$\text{LFCTNN-FC}_{\text{TGRS24}}$	34.5451	0.8906	4.8351	2.7887	3.7386	34.8682	0.9513	5.6073	2.4990	3.5248	29.1921	0.8122	3.8255	2.1648	7.0349
$\text{LFCTNN-FL}_{\text{TGRS24}}$	34.7729	0.8938	4.7992	2.7353	3.6219	35.4630	0.9567	4.9560	2.3160	3.2753	29.1034	0.8026	3.9918	2.1790	7.0973
$\text{GTNN}_{\text{TNNLS25}}$	33.5474	0.8797	5.5692	3.1067	4.1477	33.7549	0.9371	6.4975	2.8317	3.9963	28.9650	0.7949	3.8025	2.2897	7.1492
GNLTN	36.0278	0.9231	3.5060	2.3010	3.1119	35.9876	0.9585	4.5418	2.1914	3.1400	30.5365	0.8581	2.7048	1.8503	6.0137

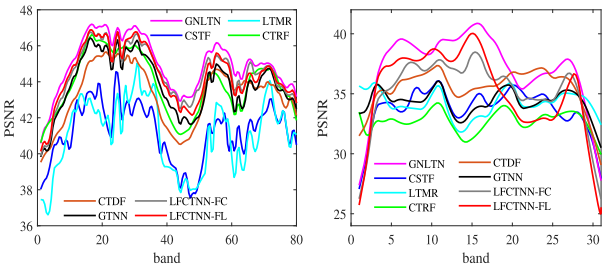


Fig. 6. PSNR values versus different bands on (a) PC and (b) *thread spools* image from the CAVE dataset.

dataset, GNLTN continues to deliver superior results across all metrics. Specifically, at SNR = 30 dB, GNLTN achieves a PSNR of 42.2872 dB, outperforming CSTF by 29.86% and GTNN by 3.13%. The method also shows a significant improvement in the SAM metric, with a 14.28% improvement over LFCTNN-FL. These findings reinforce GNLTN's capability to preserve image quality under more challenging noise conditions. Similarly, for CM, GNLTN excels in both PSNR and SAM across various noise levels. At SNR = 35 dB, GNLTN achieves a PSNR of 39.2995 dB, surpassing GTNN by a substantial margin. The SAM metric further emphasizes GNLTN's ability to capture finer image details, with a 14.82% improvement over GTNN at this noise level.

In summary, the results presented in Table III demonstrate that GNLTN consistently outperforms others across the three datasets. Notably, GNLTN maintains high-quality reconstructions in both high and low SNR conditions, demonstrating its versatility across different sampling strategies.

2) Qualitative Analysis:

a) Gaussian sampling: The CAVE *feathers* dataset, shown in Fig. 7, demonstrates that the surface reconstruction generated by the CSTF method exhibits prominent vertical striping artifacts and noise, which significantly hinder the subsequent use of the image. The LTMR, CTRF, and CTDF methods produce relatively smoother reconstructions, yet visible noise remains in the magnified regions, indicating a failure to simultaneously balance noise reduction and texture preservation. The GTNN, LFCTNN-FC, and LFCTNN-FL methods show improvements in color reproduction and detail retention; however, some blurring and loss of fine textures are still noticeable. Notably, the magnified image of LFCTNN-FL in the lower right corner reveals an inaccurate portrayal of surface gloss, particularly in highlighted areas (e.g., red), where the glossiness is not faithfully rendered. Among these methods, GNLTN stands out, providing clear images with natural gloss and well-preserved textures, demonstrating exceptional reconstruction quality.

In the *thread spools* dataset shown in Fig. 8, the CSTF method struggles significantly with fine detail reconstruction, exhibiting both horizontal and vertical artifacts in the lower left region, along with noticeable noise. While the LTMR, CTRF, CTDF, and GTNN methods demonstrate improved detail capture, their denoising capabilities still leave room for further enhancement. The LFCTNN-FC and LFCTNN-FL methods exhibit strong denoising performance; however, the lower left region is marred by unwanted artifacts, and the lower right

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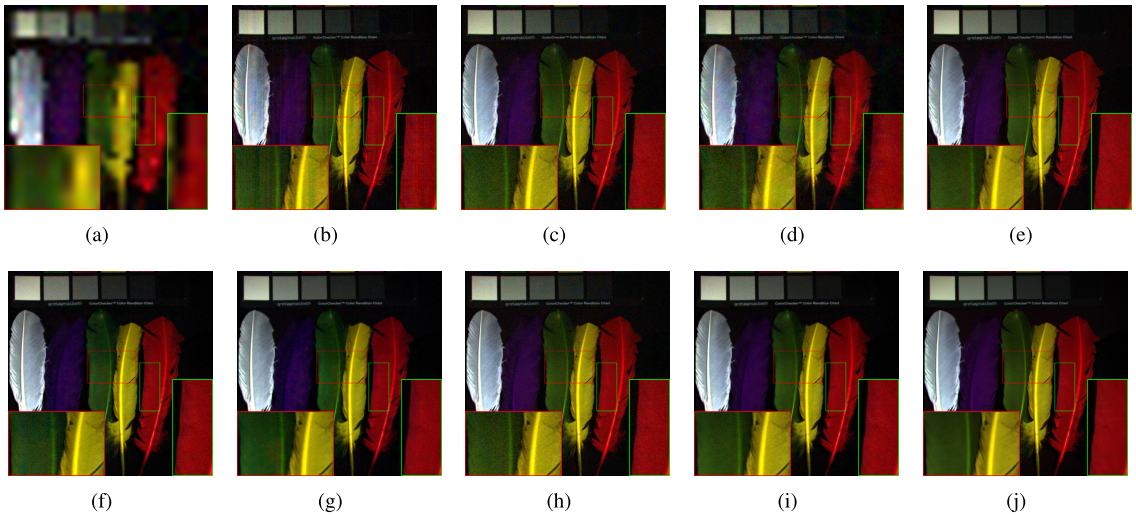


Fig. 7. Pseudo-color images (30th, 15th, and 5th bands) on the *feathers* image from the CAVE dataset. (a) LRHSI. (b) CSTF [28]. (c) LTMR [27]. (d) CTRF [45]. (e) CTDF [2]. (f) LFCTNN-FC [38]. (g) LFCTNN-FL [38]. (h) GTNN [1]. (i) GNLTN. (j) Ground truth.

region incorrectly renders purple as blue. Among all methods, GNLTN achieves near-perfect reconstruction, preserving rich details and delivering realistic color reproduction.

b) Uniform sampling: Fig. 9 presents the reconstructed pseudo-color images on PU under uniform sampling patterns, with an SNR of 20 dB for both LRHSI and HRMSI. Overall, CSTF and LTMR exhibit clear blurring artifacts, leading to the loss of fine-grained textures. While CTRF and CTDF alleviate some distortions, they still suffer from noticeable spectral inconsistencies and noise-like artifacts, as reflected by the intensified errors in the corresponding maps. LFCTNN-FC and LFCTNN-FL further improve reconstruction quality, yet residual noise and boundary artifacts remain particularly evident in regions with high-frequency structures. Although GTNN tends to reduce the overall reconstruction error, local inconsistencies and sporadic noise patterns are still present, suggesting limitations in preserving fine textures. In contrast, the reconstructed images generated by GNLTN demonstrate enhanced structural preservation, sharper edge continuity, and improved spectral fidelity.

The reconstruction results on CM under uniform sampling are as shown in Fig. 10, with an SNR of 20 dB. As observed, CSTF fails in preserving detailed spatial structures, exhibiting severe blurring and spectral distortions accompanied by extensive noise artifacts. LTMR, CTRF, and CTDF improve the sharpness to some extent; however, they still suffer from noticeable color deviations and residual noise, as indicated by the nonuniform distributions in their corresponding error maps. LFCTNN-FC and LFCTNN-FL deliver enhanced structural preservation, yet noise residuals remain discernible, particularly in regions with complex textures. GTNN achieves lower reconstruction errors and smoother spectral consistency, but local blurring and sporadic noise patches persist, especially within fine-scale structures. In contrast, the proposed GNLTN attains superior quality, preserving intricate spatial details and generating the most homogeneous and low-intensity error maps. These observations substantiate the effectiveness of GNLTN in modeling complex hyperspectral distributions and in robustly recovering high-frequency information under noisy acquisition conditions.

C. Results on Real Data

To further assess the performance of the methods, we conducted experiments on real LRHSI and HRMSI datasets, complementing the simulations performed in previous tests. In the absence of ground-truth HSI data, visual evaluation serves as the primary means to demonstrate the effectiveness of the proposed methods for real HSI super-resolution. Fig. 11 shows the super-resolved images obtained from the informed methods applied to the real dataset. Among the methods evaluated, CTDF yields the least favorable results, exhibiting severe artifacts that significantly hinder its practical applicability. While CSTF shows notable improvements, it still suffers from substantial blurring and fails to effectively capture fine spatial details. LTMR and CTRF, on the other hand, introduce noticeable jagged edges in some regions, leading to discontinuities along image boundaries. In contrast, the GTNN, LFCTNN-FC, LFCTNN-FU, and the proposed GNLTN methods excel at preserving spatial details, offering superior image quality and more coherent reconstructions.

D. Ablations and Discussions

1) Convergence Behavior: The numerical convergence of the ADMM-based solver for our model is evaluated empirically. We compute the relative error $(\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F / \|\mathcal{X}^t\|_F)$ between successively recovered images. The curve in (a) of Fig. 12 indicates that, in the Pavia dataset, the relative error decreases rapidly as the number of iterations increases, stabilizing after 20 iterations. In contrast, the relative error in the CAVE [see Fig. 12(b)] dataset exhibits initial fluctuations, but eventually approaches a stable value as the number of iterations increases, demonstrating numerical convergence. These two trends collectively validate the convergence of the proposed algorithm.

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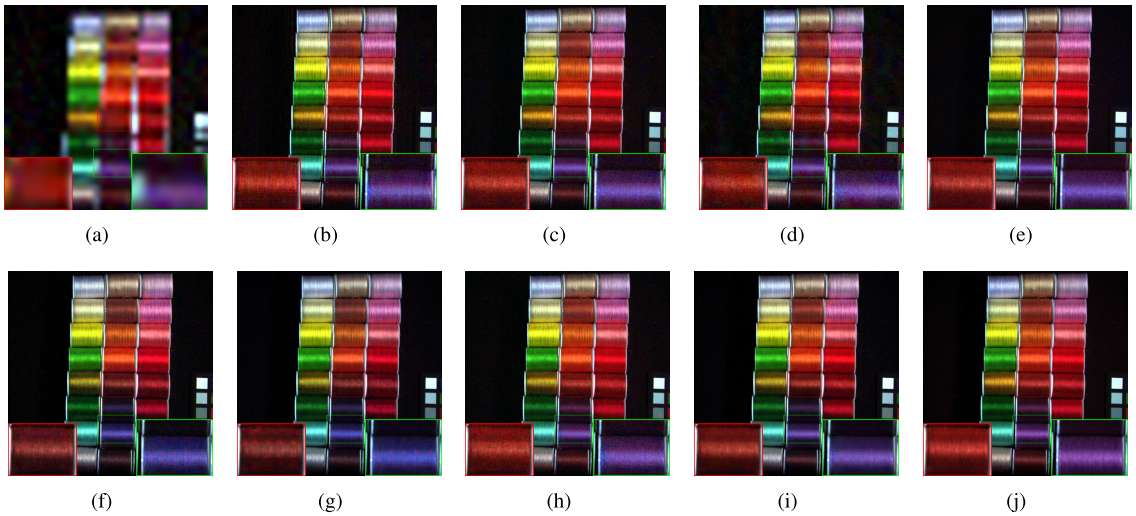


Fig. 8. Pseudo-color images (30th, 15th, and 5th bands) on *thread spools* image from the CAVE dataset. (a) LRHSI. (b) CSTF [28]. (c) LTMR [27]. (d) CTRF [45]. (e) CTDF [2]. (f) LFCTNN-FC [38]. (g) LFCTNN-FL [38]. (h) GTNN [1]. (i) GNLTN. (j) Ground truth.

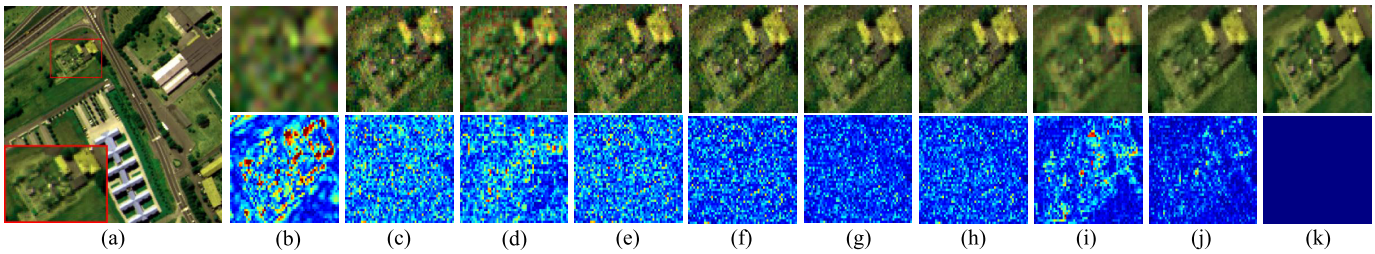


Fig. 9. Pseudo-color images (50th, 60th, and 25th bands) on the Pavia University dataset. (a) Overall image of GT. (b)–(k) Reconstructed images and corresponding error maps. The first row shows the reconstructed results obtained by different methods: (b) LRHSI. (c) CSTF [28]. (d) LTMR [27]. (e) CTRF [45]. (f) CTDF [2]. (g) LFCTNN-FC [38]. (h) LFCTNN-FL [38]. (i) GTNN [1]. (j) GNLTN. (k) GT. The second row displays the corresponding error maps between the reconstructed images and GT.

2) Analysis of Atom Selection for Parameter μ_1–μ_4: To evaluate the influence of the regularization parameters μ_1–μ_4 in (26), a sensitivity analysis is performed on PU and PC, as shown in Fig. 13. The results indicate that μ_1 and μ_3 achieve the highest PSNR values when set to 0.01 across both datasets. Although μ_2 reaches suboptimal performance at 0.01, we adopt this unified value for consistency in parameter selection. In contrast, the performance of μ_4 decreases significantly when the value exceeds 0.001, particularly on the Pavia University dataset. Given that μ_4 controls the mode-3 regularization, which aligns more closely with spectral consistency, we select $\mu_4 = 0.001$ as the default. These settings strike a balance between performance and generalization across different datasets.

3) Analysis of Atom Selection for Parameter L: The parameter L determines the extent of spatial dimension reduction. Fig. 14 illustrates the PSNR curves for the PC and cd datasets as a function of L. We observe that as L increases, the PSNR values improve significantly, reflecting enhanced preservation of image quality due to moderate dimensional compression. However, beyond $L = 8$, the improvement rate diminishes, indicating diminishing returns. Notably, when L exceeds 10, the PSNR values exhibit a slight decline, which may be attributed to excessive compression introducing information redundancy or overfitting. Based on these observations, we recommend setting L to 10 as a balanced tradeoff between performance and efficiency.

4) Influence of Low-Rank Tensor Regularization: The GNLTN method is based on the proposed low-rank tensor regularization, with the parameters λ_1, λ_2, and λ_3 being crucial, as they control the low-rank structure of the recovered tensor. Fig. 14(b) and (c) presents the PSNR curves for the PC and cd datasets across different values of λ_1, λ_2, and λ_3. Since mode-3 predominantly captures spectral information, λ_3 is set independently of $\lambda_1 = \lambda_2$. From the results shown for the PU dataset in Fig. 14(b), we observe that PSNR increases with λ_3, indicating that larger values of λ_3 amplify the spectral influence, leading to improved fusion quality. On the other hand, λ_1 and λ_2 exhibit minimal fluctuation, with PSNR peaking at 5×10^{-2}. Based on these observations, we recommend an initial setting of $[5 \times 10^{-2}, 5 \times 10^{-2}, 1 \times 10^{-2}]$. For the CAVE dataset, which contains 32 images, we set $\lambda_1 = \lambda_2 = \lambda_3$ to simplify the parameter selection. As shown in Fig. 14(c), PSNR increases sharply as λ varies from 0.1 to 2. However, when λ exceeds this peak, further increases cause a rapid decline in performance. Based on these results, we recommend setting $\lambda = 2$ for the CAVE dataset.

5) Results on Running Time: Computational efficiency is a crucial metric for the practical application of newly proposed models. Fig. 15 presents the computational times of all tested

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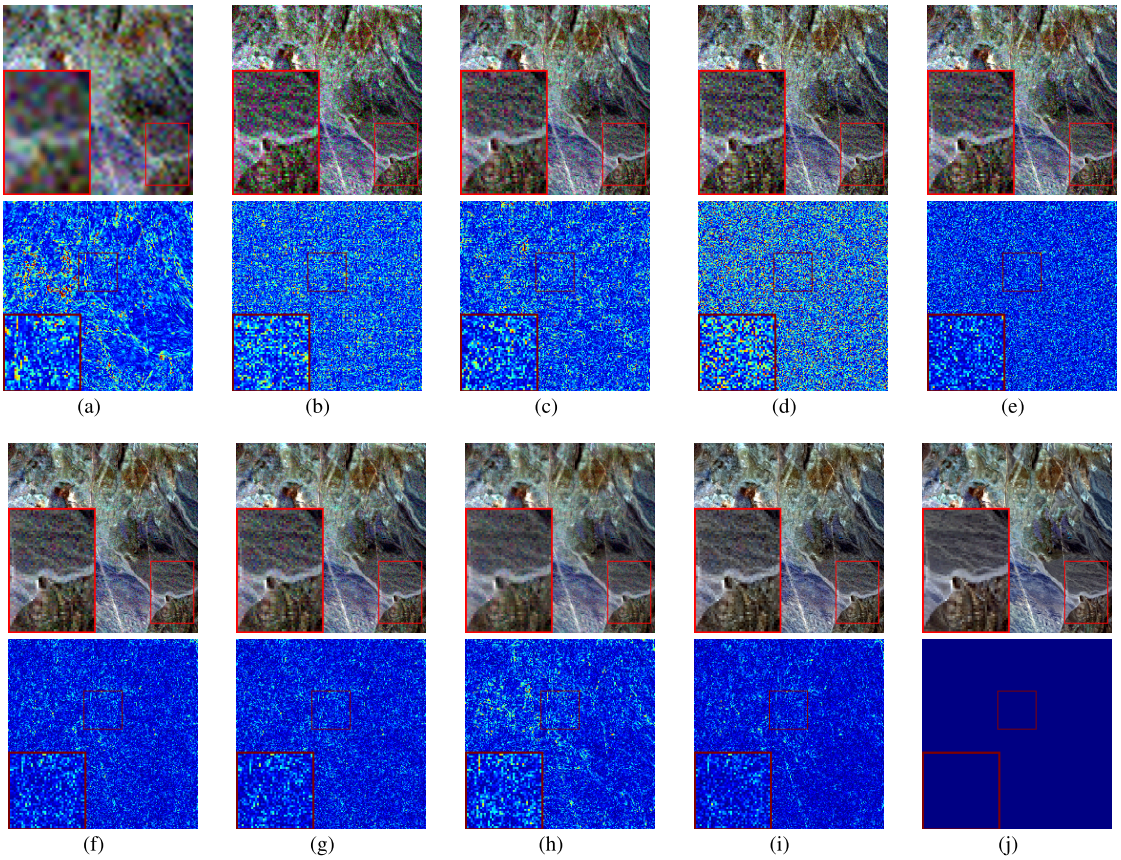


Fig. 10. Pseudo-color images (19th, 32th, and 52th bands) and the error images on Cuprite mine dataset. (a) LRHSI. (b) CSTF [28]. (c) LTMR [27]. (d) CTRF [45]. (e) CTDF [2]. (f) LFCTNN-FC [38]. (g) LFCTNN-FL [38]. (h) GTNN [1]. (i) GNLTN. (j) Ground truth.



Fig. 11. Pseudo-color images (10th, 9th, and 1st bands) in real HSI and MSI fusion. (a) LRHSI. (b) CSTF [28]. (c) LTMR [27]. (d) CTRF [45]. (e) CTDF [2]. (f) LFCTNN-FC [38]. (g) LFCTNN-FL [38]. (h) GTNN [1]. (i) GNLTN.

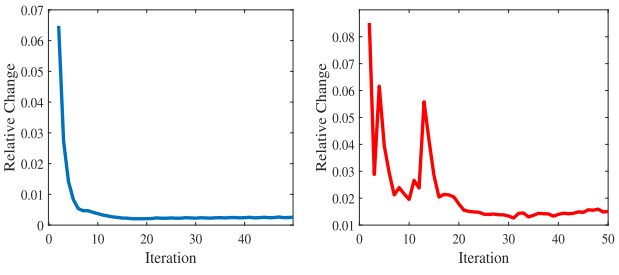


Fig. 12. Relative error of $(\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F / \|\mathcal{X}^t\|_F)$ versus increasing iterations of (a) PC dataset and (b) cd of CAVE dataset.

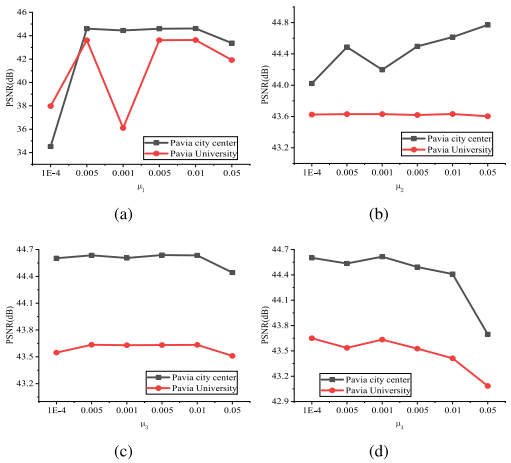
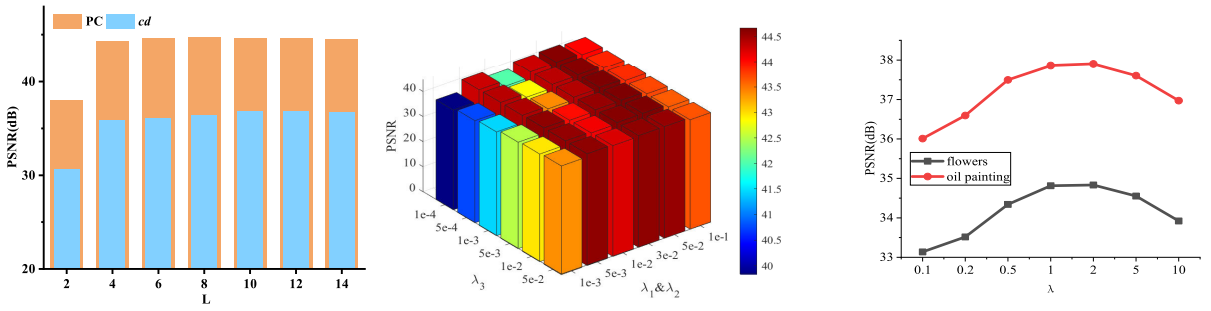


Fig. 13. PSNR performance with varying values of (a) μ_1, (b) μ_2, (c) μ_3, and (d) μ_4 on the Pavia University and Pavia city center datasets.

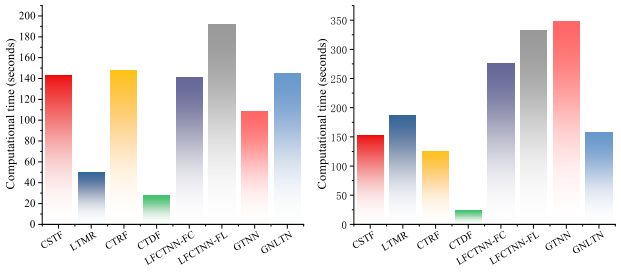
methods on the PU and cd datasets. On the PU dataset, CTDF and LTMR exhibit the highest computational efficiency among all competing methods, primarily due to their streamlined model architectures. In contrast, LFCTNN, GTNN, and the proposed GNLTN, which involve low-rank exploration across the three modes of the input tensor, demonstrate a tradeoff between efficiency and performance. On the *CAVE* dataset, the

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(a) (b) (c)

Fig. 14. PSNR values for different selections of (a) subspace dimension, (b) $\lambda_{1,2,3}$ on the PU dataset, and (c) $\lambda_{1,2,3}$ on the cd dataset.

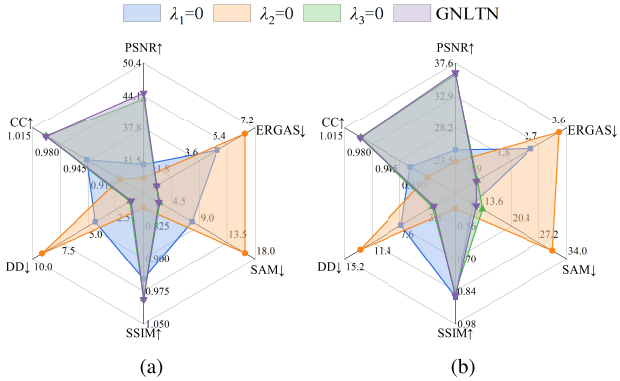


(a) (b)

Fig. 15. Computational time (in seconds) of all test methods on (a) PU dataset and (b) cd image from the CAVE dataset.

TABLE IV
ABLATION EXPERIMENTS ON THE PU AND CD DATASETS FOR DOMAIN TRANSFORM

Nonlinear	DFT	Self-learning	PU	*cd*
		✔	40.6326	34.6459
	✔		42.9282	32.5116
	✔	✔	43.4734	35.6291
✔	✔		43.4257	34.7108
✔		✔	40.7167	33.6015
✔	✔	✔	43.6317	36.2321



(a) (b)

Fig. 16. Ablation experiments on (a) PC and (b) cd dataset for the full-scale architecture.

algorithmic runtime increases significantly with the growing spatial dimensions, with the proposed GNLTN achieving superior fusion in fewer iterations due to its nonlinear transform mechanism. Overall, while GNLTN's computational efficiency is moderate, the relatively higher performance justifies the incurred runtime.

6) Roles of NLT-Domain: Domain transform plays a critical role in capturing additional low-rank characteristics. Table IV presents a comparison of ablation results across different transform domains for the PU and cd datasets. Our findings reveal that utilizing individual domains—specifically DFT and self-learning—results in substantial performance degradation in both datasets. This is because these individual approaches fail to sufficiently explore the low-rank space of HSI. In contrast, the combined domain approach yields superior performance, as it leverages the strengths of both DFT and self-learning to better capture and represent HSI's underlying structure. Similarly, when guided by a nonlinear transform domain, both DFT and self-learning, as standalone methods, are still unable to effectively explore the low-rank characteristics of the data. The integration of all three approaches ensures efficient and comprehensive exploration of both linear and nonlinear features, thereby optimizing the fusion performance and yielding the best results.

7) Roles of Full-scale Architecture: To assess the contribution of each component in the full-scale model, we conducted an ablation study by sequentially deactivating individual components and evaluating their impact on overall performance. The quantitative results for the WDC and cd datasets are presented in the radar charts in Fig. 16. In these charts, "$\lambda_1 = 0$" indicates that all components are active except for the first mode of the GNLTN. As shown in the figure, removing both the first and second modes leads to a significant performance drop, while the enhancement provided by the third mode alone is relatively limited. Overall, the results highlight the critical role of all three modes in effectively capturing low-rank features, thus validating the effectiveness of the proposed method.

8) Comparison of Different Nonlinear Transforms: To examine the effect of activation function choice, comparative experiments are conducted by substituting Tanh with several commonly used ones, including ReLU, Softmax, Sigmoid, and Hard Sigmoid. The quantitative results, summarized in Table V, show that Tanh achieves consistently higher PSNR and SSIM values and lower SAM values across PU and PC. Relative to Sigmoid and Hard Sigmoid, Tanh improves the PSNR by approximately 0.5 dB and reduces the SAM by about 0.18. The observed differences can be attributed to the

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TABLE V
COMPARISON OF DIFFERENT ACTIVATION FUNCTIONS ON PU AND PC

Methods	PU			PC		
	PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓
ReLU	34.7076	0.9345	3.6873	34.3179	0.9444	5.2671
Softmax	38.0562	0.9508	4.0484	38.1070	0.9713	5.6902
Sigmoid	43.0953	0.9859	2.0337	43.5159	0.9923	2.7821
Hard Sigmoid	43.0936	0.9859	2.0336	43.5172	0.9924	2.7804
Tanh	**43.6317**	**0.9875**	**1.8549**	**44.6151**	**0.9939**	**2.4438**

intrinsic properties of the activation functions. ReLU introduces nonsmoothness at zero and produces sparse activations, which may limit the modeling of subtle spectral variations. Softmax, primarily designed for classification, is less effective for continuous-valued regression tasks such as hyperspectral image reconstruction. Sigmoid and Hard Sigmoid provide smoother mappings but are prone to saturation at extreme input values, thereby restricting gradient propagation. Tanh, characterized by a symmetric, smooth, and nonsaturating mapping around zero, facilitates stable gradient flow and better captures the nonlinear relationships within the data.

VI. CONCLUSION

In this work, we address the limitations of conventional TNN-based HSISR methods by introducing the NLT-domain transform framework. This approach integrates nonlinear, DFT, and self-learning transforms, effectively enhancing low-rank approximation by promoting singular value concentration and refining rank estimation accuracy. Building upon this framework, we develop a logarithmic-based, NLT-transform-oriented TNN (GNLTN), designed to mitigate singular value over-penalization and improve robustness across multiple tensor modes. To further exploit spatial–spectral structures, our method incorporates spectral subspace estimation and clustering-based patch grouping, allowing for a more accurate representation of spectral and spatial relationships. Extensive experimental evaluations demonstrate that our approach significantly outperforms existing HSISR methods in both reconstruction accuracy and visual fidelity. These findings underscore the effectiveness of our proposed framework in advancing HSISR, offering a robust and generalizable solution to the challenges of hyperspectral image reconstruction.

REFERENCES

[1] R. Dian, Y. Liu, and S. Li, "Hyperspectral image fusion via a novel generalized tensor nuclear norm regularization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7437–7448, Apr. 2025.

[2] T. Xu et al., "A coupled tensor double-factor method for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515417.

[3] Y. Bu, Y. Zhao, J. Xue, J. Yao, and J. C.-W. Chan, "Transferable multiple subspace learning for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[4] C. Zhu, S. Deng, X. Song, Y. Li, and Q. Wang, "Mamba collaborative implicit neural representation for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515417.

[5] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[6] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6007305.

[7] C. Zhu, S. Deng, Y. Zhou, L.-J. Deng, and Q. Wu, "QIS-GAN: A lightweight adversarial network with quadtree implicit sampling for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531115.

[8] K. Maurya and S. Mahajan, "Mangrove species discrimination using modified spectral angle mapper algorithm on hyperspectral imagery: A study at marine national park (MNP) Jamnagar, Gulf of Kutch, Gujarat and lothian Island, Sundarbans, West Bengal," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121455.

[9] Y. Gu, S. Chen, X. Sun, J. Ji, Y. Zhou, and R. Ji, "Optical remote sensing image salient object detection via bidirectional cross-attention and attention restoration," *Pattern Recognit.*, vol. 164, Aug. 2025, Art. no. 111478.

[10] Y. Quan, H. Xu, R. Wang, Q. Guan, and J. Zheng, "ORSI salient object detection via progressive semantic flow and uncertainty-aware refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608013.

[11] Y. Gu, H. Xu, Y. Quan, W. Chen, and J. Zheng, "ORSI salient object detection via bidimensional attention and full-stage semantic guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603213.

[12] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[13] J. Zheng, P. Li, H. Xu, J. Jiang, Y. Feng, and Z. Liu, "Hyperspectral image superresolution via subspace-based deep prior regularization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2437–2449, 2023.

[14] W. Wang, L.-J. Deng, R. Ran, and G. Vivone, "A general paradigm with detail-preserving conditional invertible network for image fusion," *Int. J. Comput. Vis.*, vol. 132, no. 4, pp. 1029–1054, Apr. 2024.

[15] Q. Zhu, M. Zhang, Y. Chen, G. Zheng, and J. Luo, "Spectral correlation-based fusion network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5500314.

[16] Q. Hu, X. Wang, J. Jiang, X.-P. Zhang, and J. Ma, "Exploring the spectral prior for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5260–5272, 2024.

[17] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[18] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[19] X. Qi et al., "A multistage semi-supervised network for hyperspectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510017.

[20] W. Dong, J. Qu, S. Xiao, T. Zhang, Y. Li, and X. Jia, "Noise prior knowledge informed Bayesian inference network for hyperspectral super-resolution," *IEEE Trans. Image Process.*, vol. 32, pp. 3121–3135, 2023.

[21] J. Qu et al., "Progressive multi-iteration registration-fusion co-optimization network for unregistered hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519814.

[22] Y. Wang, W. Li, Y. Gui, Q. Du, and J. E. Fowler, "A generalized tensor formulation for hyperspectral image super-resolution under general spatial blurring," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4684–4698, Jun. 2025, doi: 10.1109/TPAMI.2025.3545605.

[23] J. Long et al., "Parameter-free spectral–spatial optimization algorithm for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5617415.

[24] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, T.-H. Ma, and T.-Y. Ji, "Mixed noise removal in hyperspectral image via low-fibered-rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 734–749, Jan. 2020.

[25] W. Qin, H. Wang, F. Zhang, W. Ma, J. Wang, and T. Huang, "Nonconvex robust high-order tensor completion using randomized low-rank approximation," *IEEE Trans. Image Process.*, vol. 33, pp. 2835–2850, 2024.

[26] Y. Bu, Y. Zhao, and J. C.-W. Chan, "Mixed norm regularized models for low-rank tensor completion," *Inf. Sci.*, vol. 670, Jun. 2024, Art. no. 120630.

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[27] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[28] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[29] H. Xu, M. Qin, Y. Yan, M. Zhang, and J. Zheng, "Nonlocal B-spline representation of tensor decomposition for hyperspectral image inpainting," *Signal Process.*, vol. 206, May 2023, Art. no. 108888.

[30] F. Ma, Q. Qu, F. Yang, and G. Xu, "Hankel tensor subspace representation for remotely sensed image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 8630–8642, 2025.

[31] Z.-C. Wu, Y.-J. Li, T.-Z. Huang, L.-J. Deng, and G. Vivone, "CroDoSR: Tensor cross-domain rank for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530315.

[32] C. Lu, "Transforms based tensor robust PCA: Corrupted low-rank tensors recovery via convex optimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1145–1152.

[33] G. Song, M. K. Ng, and X. Zhang, "Robust tensor completion using transformed tensor singular value decomposition," *Numer. Linear Algebra Appl.*, vol. 27, no. 3, p. 2299, May 2020.

[34] J. Gao, H. Wang, and J. Peng, "Hyperspectral image restoration via RPCA model based on spectral–spatial correlated total variation regularizer," *Neurocomputing*, vol. 619, Feb. 2025, Art. no. 128885.

[35] T. Jiang, M. K. Ng, X. Zhao, and T. Huang, "Framelet representation of tensor nuclear norm for third-order tensor completion," *IEEE Trans. Image Process.*, vol. 29, pp. 7233–7244, 2020.

[36] T.-X. Jiang, X.-L. Zhao, H. Zhang, and M. K. Ng, "Dictionary learning with low-rank coding coefficients for tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 2, pp. 932–946, Feb. 2023.

[37] B.-Z. Li, X.-L. Zhao, T.-Y. Ji, X.-J. Zhang, and T.-Z. Huang, "Nonlinear transform induced tensor nuclear norm for tensor completion," *J. Sci. Comput.*, vol. 92, no. 3, p. 83, Sep. 2022.

[38] H. Xu, C. Fang, Y. Ge, Y. Gu, and J. Zheng, "Cascade-transform-based tensor nuclear norm for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540616.

[39] H. Xu, C. Fang, R. Wang, S. Chen, and J. Zheng, "Dual-enhanced high-order self-learning tensor singular value decomposition for robust principal component analysis," *IEEE Trans. Artif. Intell.*, vol. 5, no. 7, pp. 3564–3578, Jul. 2024.

[40] H. Xu, Y. Quan, C. Fang, and J. Zheng, "Robust principal component analysis via high-order self-learning transform tensor nuclear norm," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2024, pp. 1–6.

[41] H. Wang, J. Peng, W. Qin, J. Wang, and D. Meng, "Guaranteed tensor recovery fused low-rankness and smoothness," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10990–11007, Sep. 2023.

[42] H. Xu, M. Qin, S. Chen, Y. Zheng, and J. Zheng, "Hyperspectral-multispectral image fusion via tensor ring and subspace decompositions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8823–8837, 2021.

[43] N. Bansal, X. Chen, and Z. Wang, "Can we gain more from orthogonality regularizations in training deep networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–12.

[44] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[45] W. He, Y. Chen, N. Yokoya, C. Li, and Q. Zhao, "Hyperspectral super-resolution via coupled tensor ring factorization," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108280.

[46] H. Xu, J. Jiang, Y. Feng, Y. Jin, and J. Zheng, "Tensor completion via hybrid shallow-and-deep priors," *Int. J. Speech Technol.*, vol. 53, no. 13, pp. 17093–17114, Jul. 2023.

[47] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[48] F. Dell'Acqua, P. Gamba, A. Ferrari, J. A. Palmason, J. A. Benediktsson, and K. Árnason, "Exploiting spectral and spatial information in hyperspectral urban data with high resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 322–326, Oct. 2004.

[49] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.

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